



UNIVERSITÀ DEGLI STUDI DI SALERNO
DIPARTIMENTO DI INFORMATICA

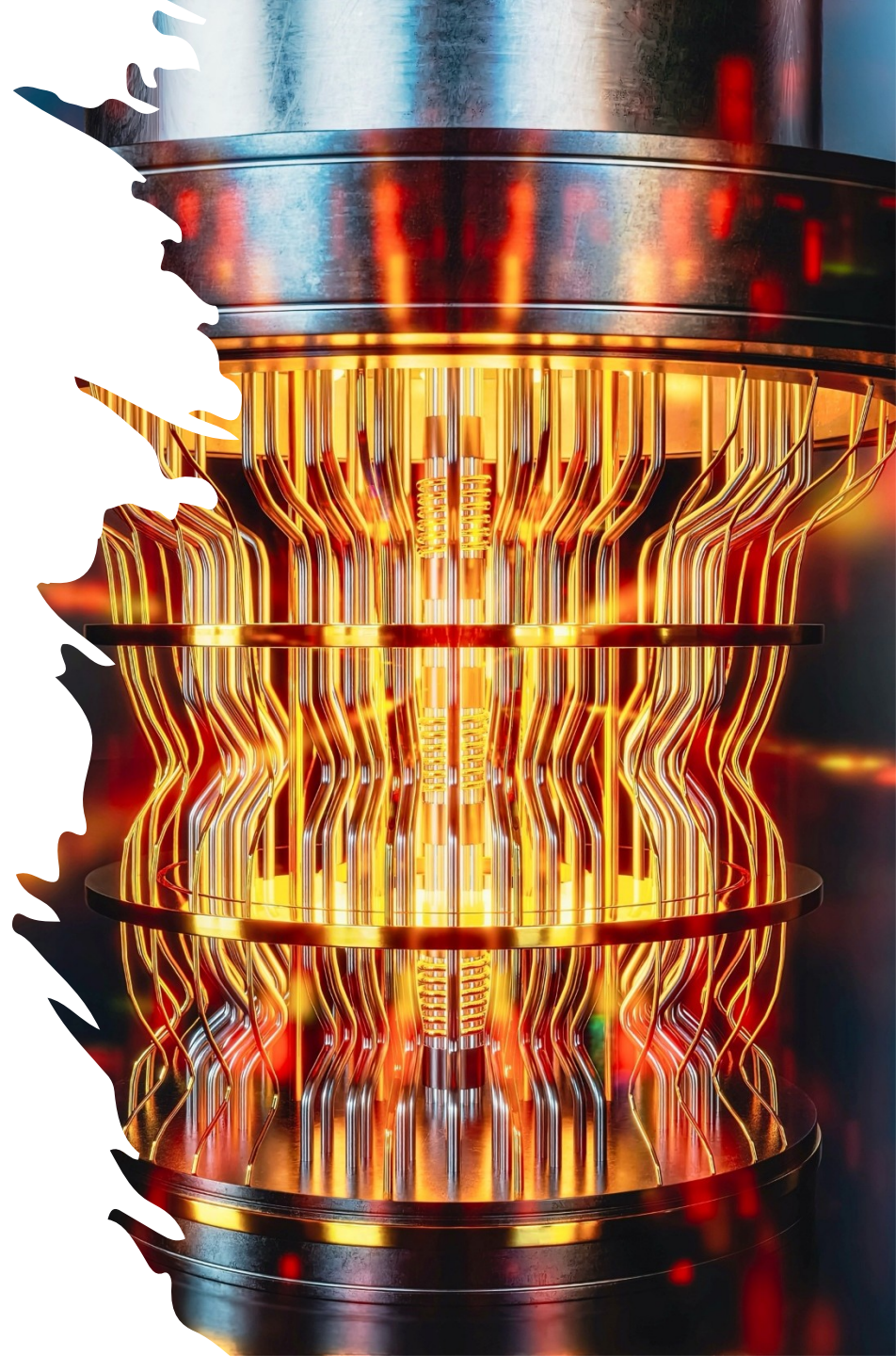
Course on Quantum Tech- nology for Security



Lecture 3 – Advanced Aspects of Quantum Computing

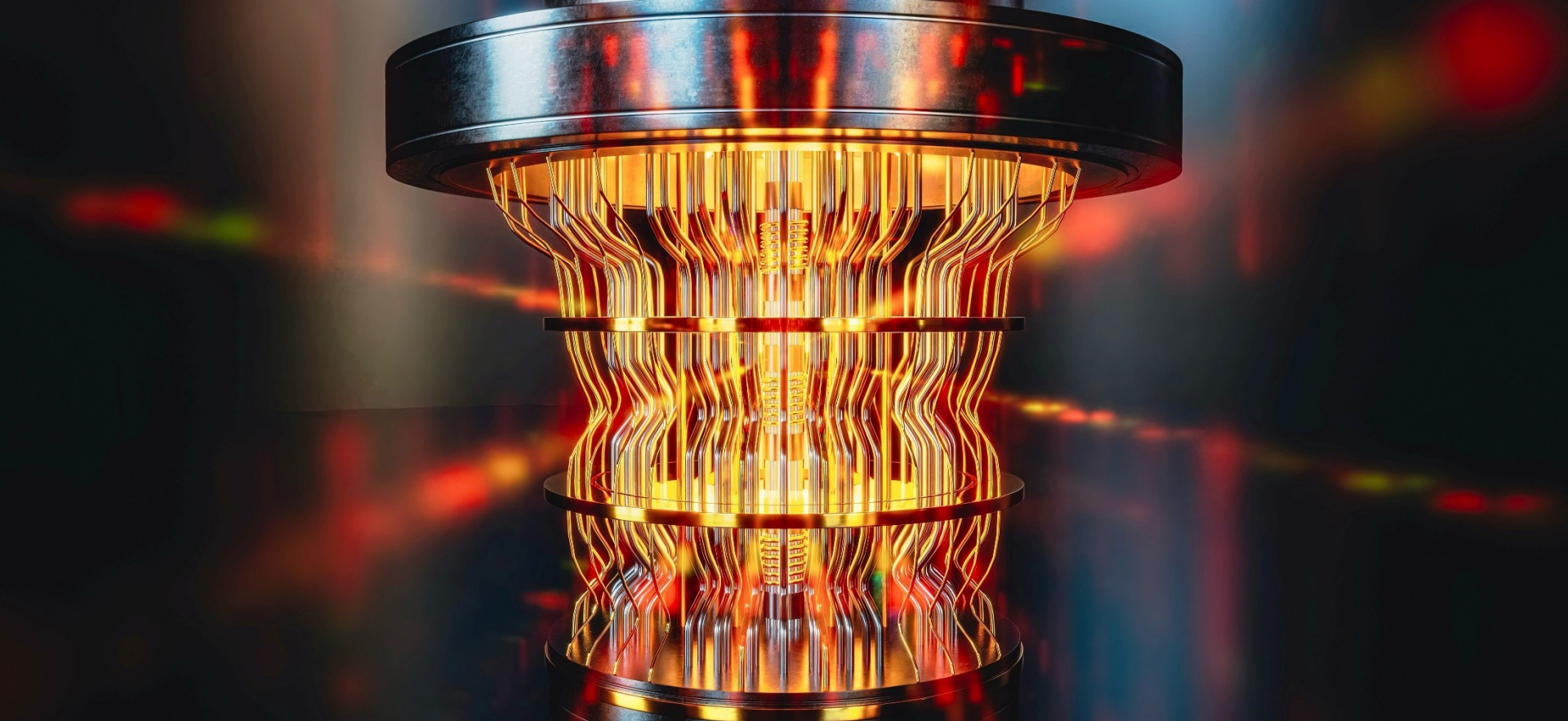


Prof. Esposito Christian



::... Summary

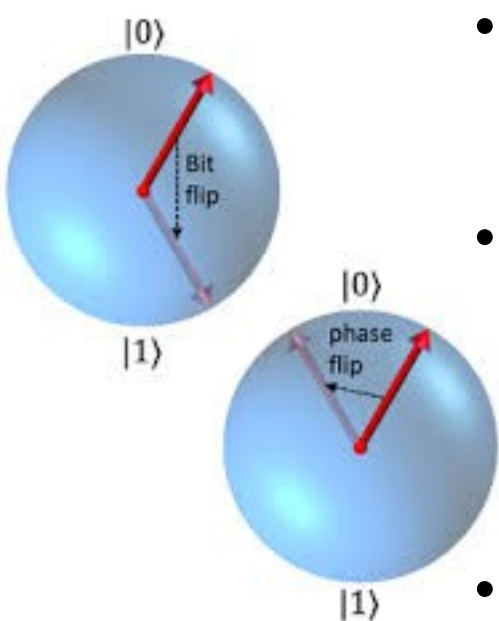
- Error Correction in Quantum Computing
 - Introduction, and Three-qubit bit and phase flip code
- Analog Programming with Hamiltonians
 - Introduction to Hamiltonians, state evolution in time, Quantum Annealing
- Quantum Optimization
 - Variational Quantum Eigensolver, Quantum Approximate Optimization Algorithm, Variational Quantum Solver
- Quantum Machine Learning
 - xxx



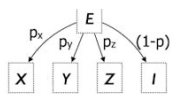
Error Correction in Quantum Computing

::: Introduction

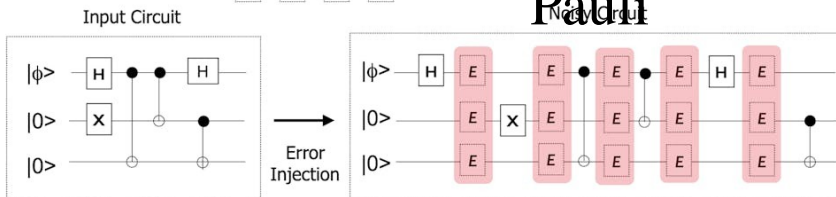
A very important issue in Quantum Computing is Error Correction, as the quantum state being prepared and manipulated in a quantum device is very delicate and easily disturbed.



- Bit-flip errors: A qubit accidentally changes from $|0\rangle$ to $|1\rangle$, or from $|1\rangle$ to $|0\rangle$. This is similar to a classical bit being flipped by noise.
- Phase-flip errors: The qubit's state keeps the same probabilities, but its phase changes, which can disrupt interference effects. These errors are uniquely important in quantum systems because phase is part of the information.
- Depolarizing errors: The qubit becomes partially randomized by applying random

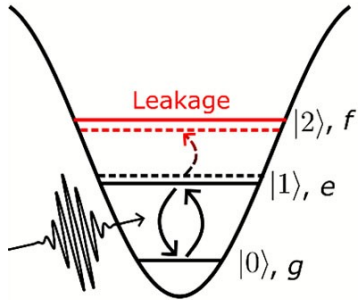


Pauli



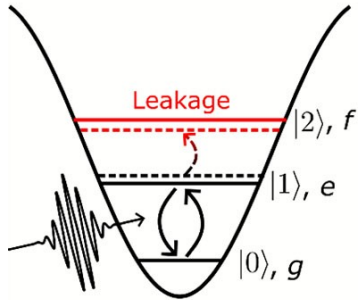
operations to a qubit, causing decoherence.

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- Leakage errors: The qubit leaves the two-level computational space used for computation and moves into some unwanted higher-energy state.
- Measurement errors: The device reports the wrong outcome when reading a qubit. For example, the qubit may really be in $|0\rangle$, but the measurement says $|1\rangle$.

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To deal with random disturbances to the quantum state, caused by noise, the solution is to add redundancy. If the state of some qubits is destroyed, it is possible to use the redundant information to reconstruct it.

Such **Error Correction** (ER) algorithms are complicated by challenges peculiar to quantum physics and not available in the classic case:

- No cloning theorem - we cannot simply copy the state of a qubit, which is the traditional redundancy manner and is ruled out.
- Measurements alter the quantum state - if a qubit is received and it has been disturbed, a simple measurement cannot be done to find out what kind of error has occurred as the quantum information is destroyed.

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- Continuum of transformations - in the classic case, the bit can only be flipped, in the quantum case, the noise can consist of an arbitrary unitary transformation or more than one, which can do anything and end at any point in the Bloch sphere.

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In the current literature, highly sophisticated Quantum EC (QEC) algorithms can address all three challenges. Let's start with a simple example called the **Three-qubit bit flip code**, where we have to deal with one type of error only: a random flip of the qubit state or with probability (assumed small). This type of error is equivalent to the random application of the Pauli X-gate with probability .

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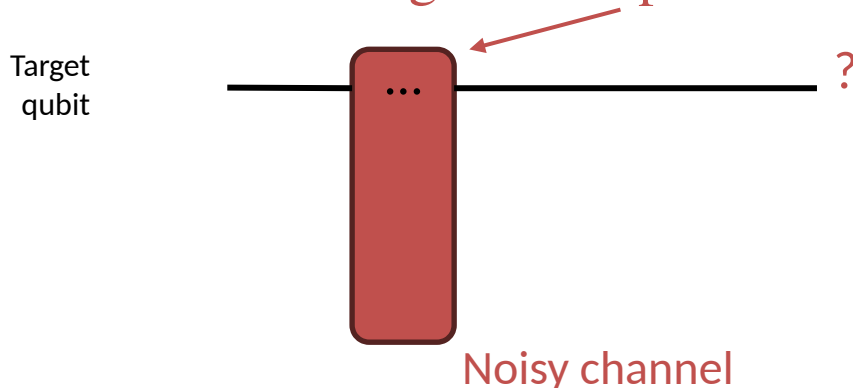
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The qubit to preserve is in the arbitrary superposition state

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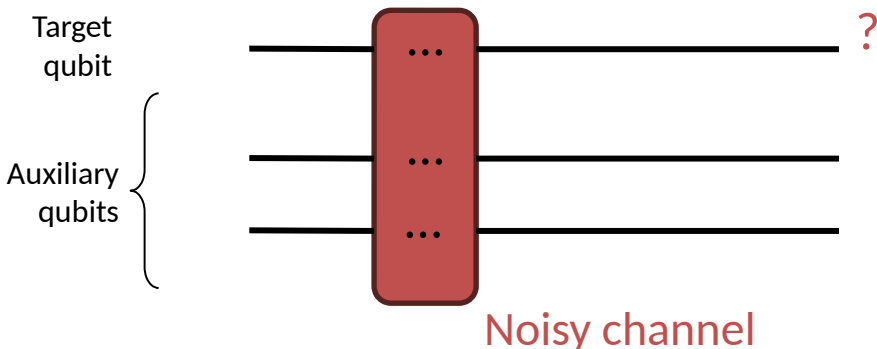


The qubit to preserve is in the arbitrary superposition state that is transmitted through a noise channel. As a consequence we do not know which state comes out.

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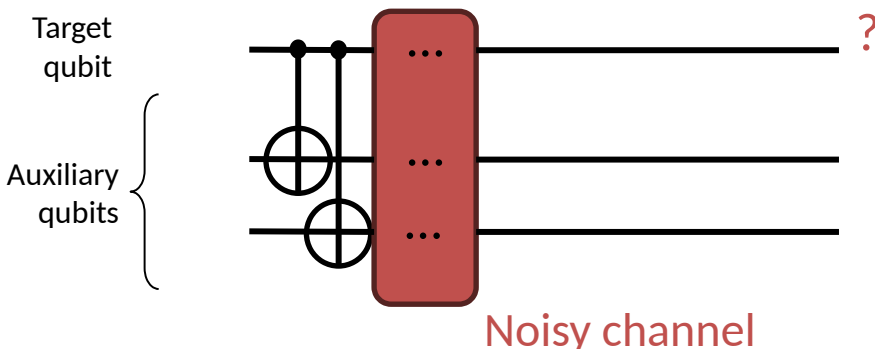


The idea of the QEC is to use two auxiliary qubits initially set to basis and used to build up a circuit for correction.

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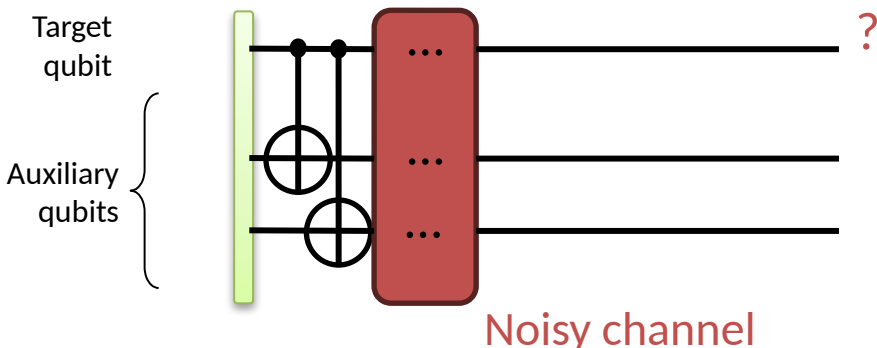
These auxiliary qubits are coupled with the target qubit via a sequence of C-NOT gates and see how the state evolves.

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$$|\psi_0\rangle = |\psi\rangle |0\rangle |0\rangle = \alpha |000\rangle + \beta |100\rangle$$

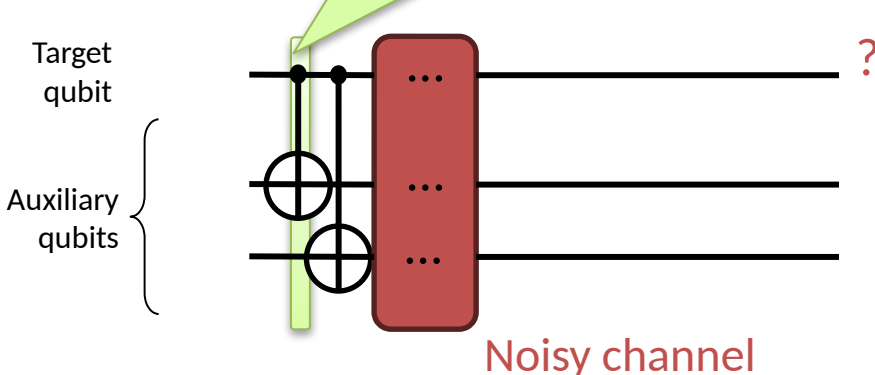


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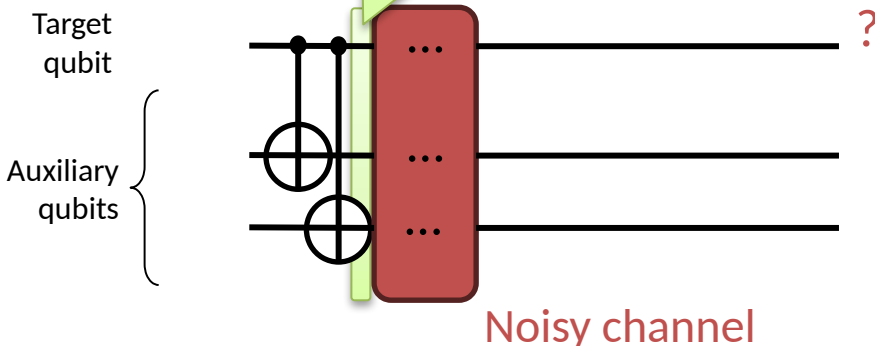


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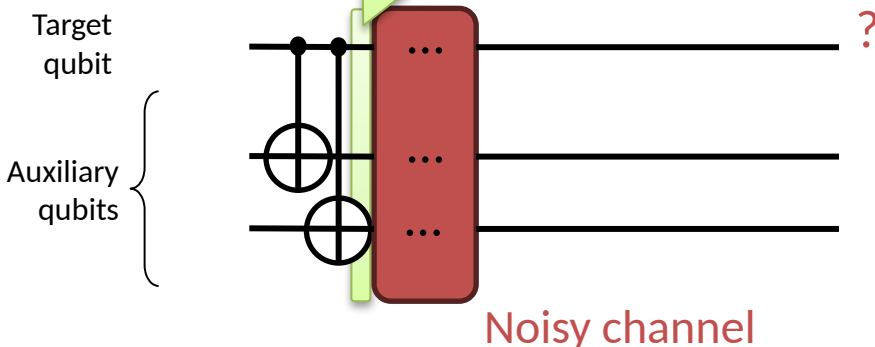


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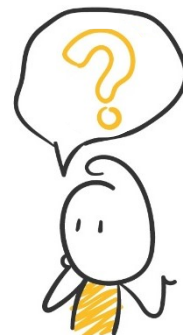
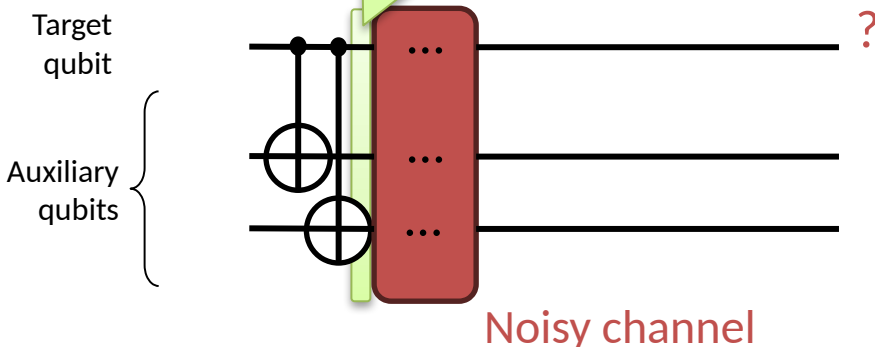


These auxiliary qubits are coupled with the target qubit via a sequence of C-NOT gates and see how the state evolves. The three qubits enter the noisy channel.

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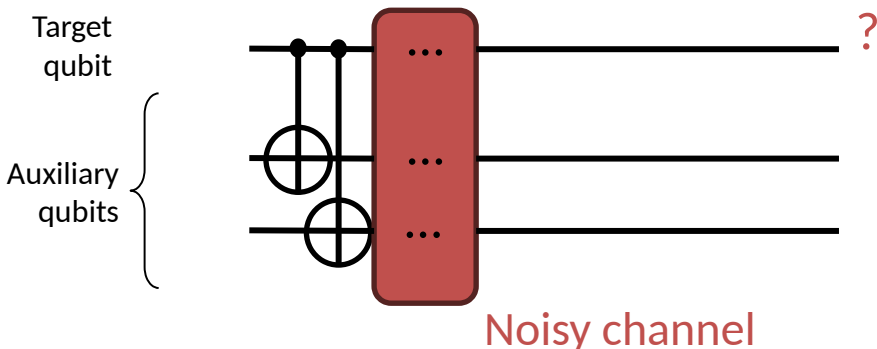


What is the state of the three qubits after the noisy channel?

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The random application of the Pauli-X gate might occur on all of these qubits, individually, but it is so unlikely two or more qubits are disturbed simultaneously (and are negligible).

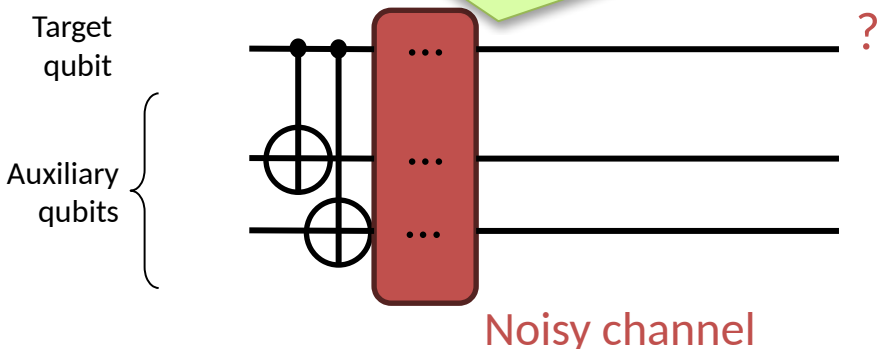
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In the current literature, highly sophisticated Quantum EC (QEC) We consider only 4 cases:

1. all of them exit the noisy channel undisturbed and ;
2. the random application of Pauli-X occurs on the first qubit: ;
3. the random application of Pauli-X occurs on the first qubit: ;
4. the random application of Pauli-X occurs on the first qubit:

of the Pauli X gate probability .

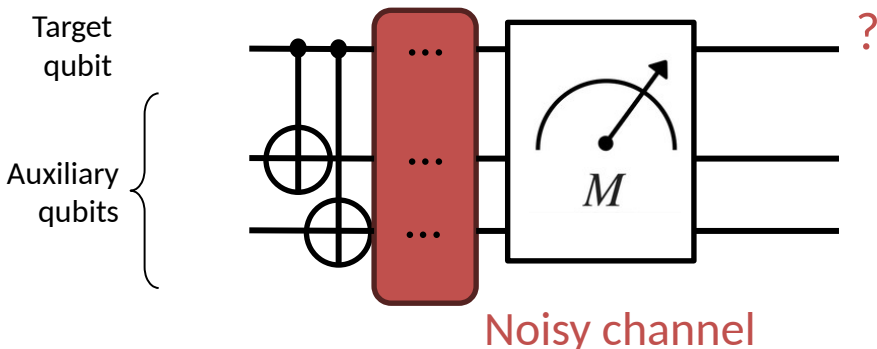


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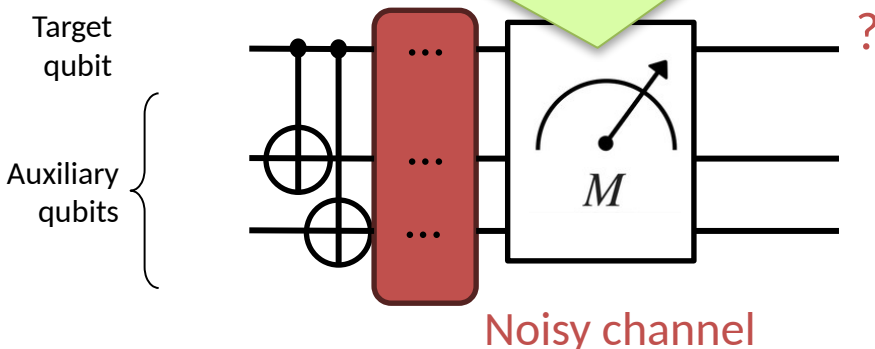
Then, the measurement is performed on all the qubits simultaneously, designed to not destroy the state of the three qubits no matter in which of the four state they are.

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The measurement is described in the 8-dimensional Hilbert Space of the 3-qubits. Each outcome of a measurement corresponds to a subspace of the Hilbert space. This measurement has 4 possible outcomes, each corresponding to a 2-dimensional subspace of the 8-dimensional Hilbert Space and are mutually orthogonal.

1. is the projector associated to the subspace defined by basis 000 and 111;
2. is the projector associated to the subspace defined by basis 100 and 011;
3. is the projector associated to the subspace defined by basis 010 and 101;
4. is the projector associated to the subspace defined by basis 001 and 110.

of the Pauli A ... ability .



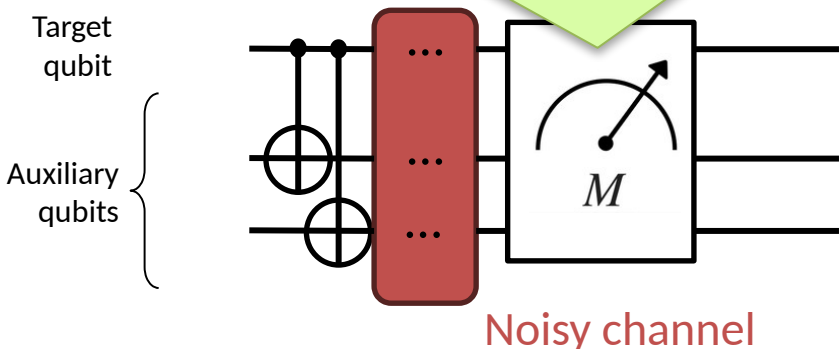
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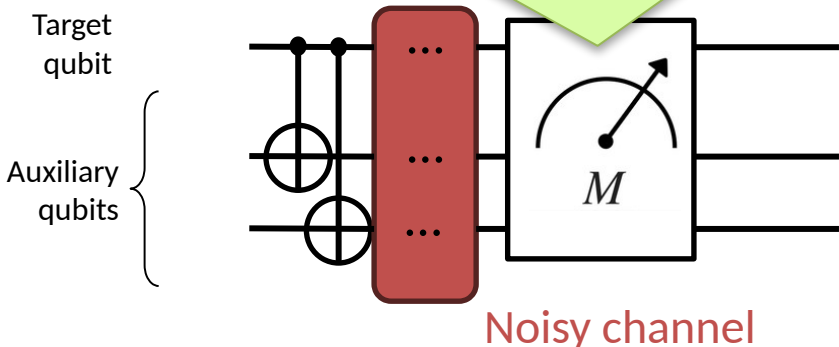
? If we remind possible states, the first one lies in the subspace defined by 000 and 111, and the measurement will give outcome 0 and since they were in that subspace they are unchanged.

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of the Pauli X and Y stability .



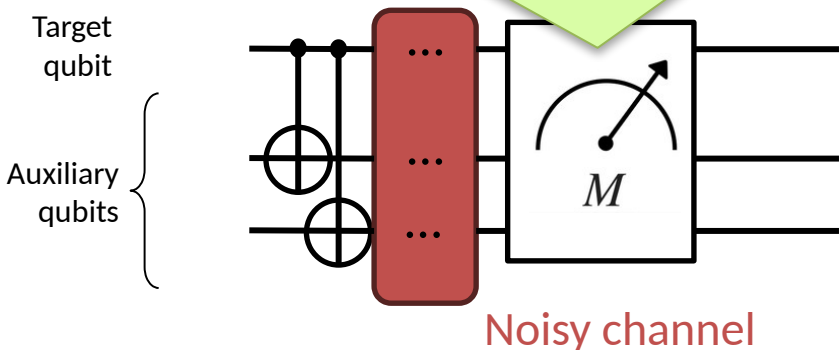
? If we remind possible states, the second one lies in the subspace defined by 100 and 011, and the measurement will give outcome 1 and since they were in that subspace they are unchanged.

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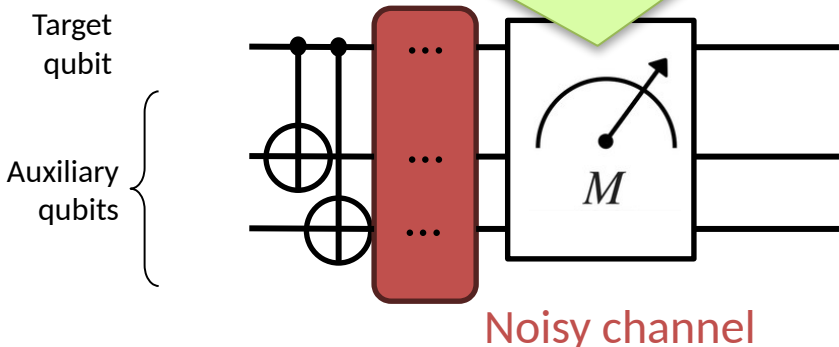
? This same applied with the third and forth possibilities, with lies in the third and forth subspaces respectively, and the measurement will give outcome 2 or 3 and they are unchanged.

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? This measurement informs in which of the four cases we are: if the disturbance occurred and at which qubit of the three. This is called syndrome measurement and returns the type of error occurred.

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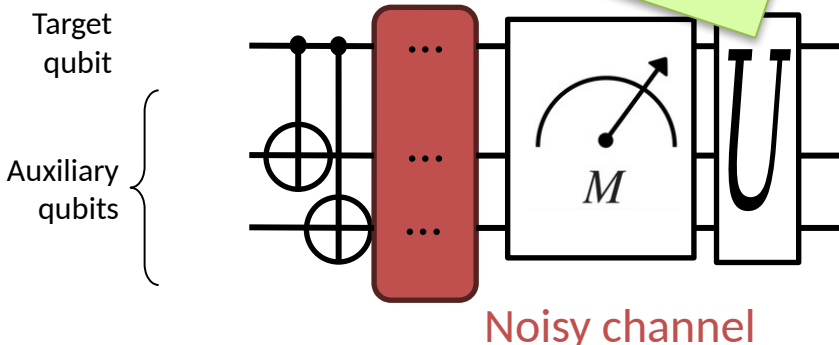
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Recovery operation: P_1, P_2, P_3

Recovery operation:

- Nothing as to be done so ;
- Only the first qubit has been flipped so another Pauli-X has to be applied to only the first qubit to undo the error;
- Only the second qubit has been flipped so another Pauli-X has to be applied to only the second qubit to undo the error;
- Only the third qubit has been flipped so another Pauli-X has to be applied to only the third qubit to undo the error.

of the Pauli X-gate reliability .

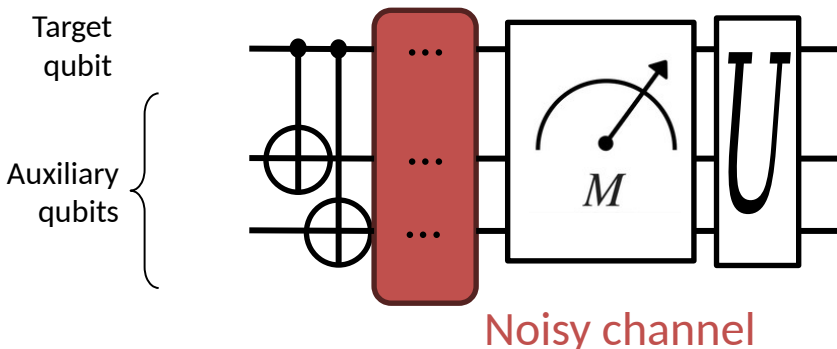


? Now the error can be corrected by applying a unitary transformation applied depending on the outcome of the syndrome measurement.

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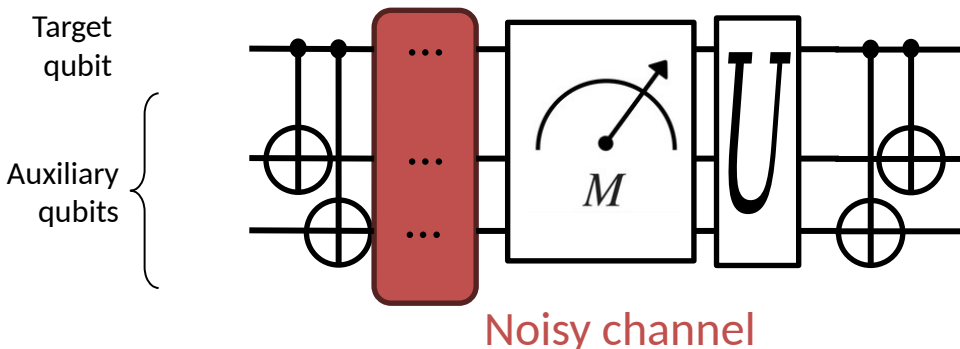


At the end the three qubits are in the initial superposition state before entering the noisy channel: .

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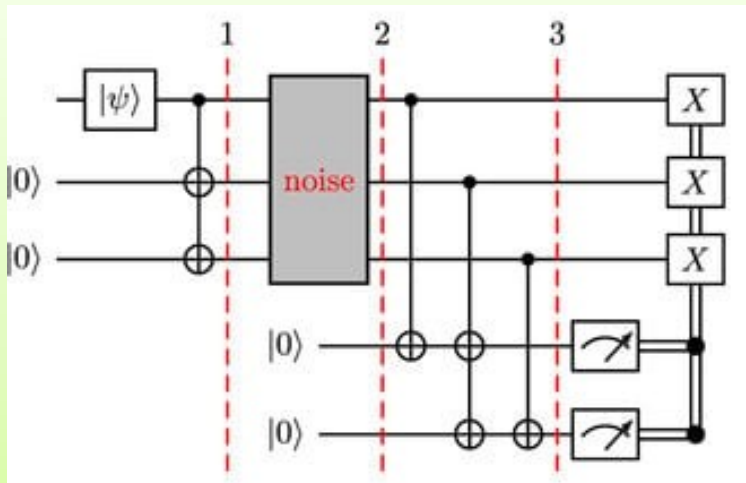
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The last step consists to apply C-NOTs again to undo the one at the beginning that created entanglement between the target and auxilliary qubits.

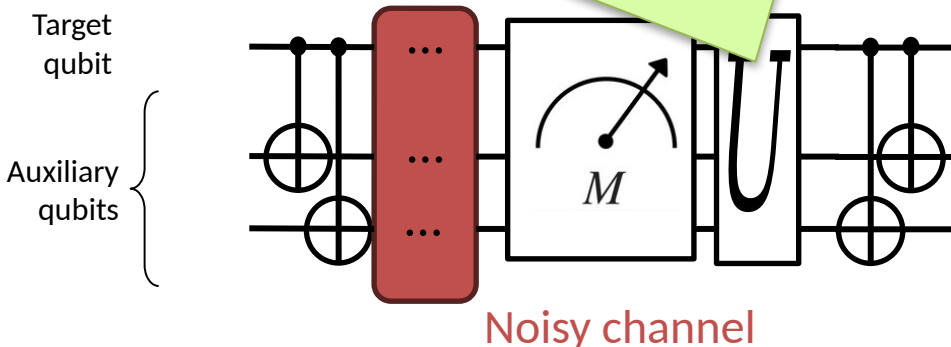
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Concrete Circuit realization: Two additional auxiliary qubits are used and measured to implement syndrome measurement, and to control the unitary recovery transformation made of controlled Pauli-X gates. After the recovery C-NOTs are reapplied.

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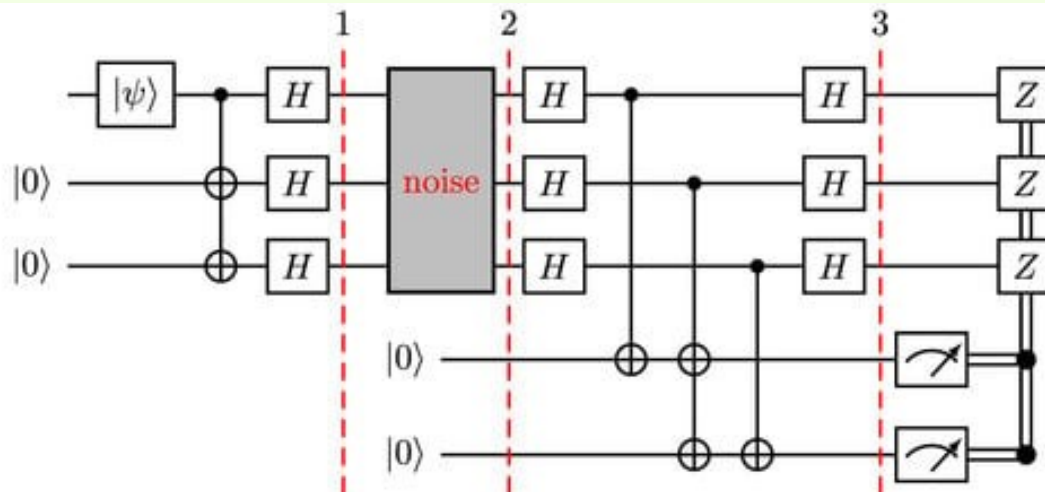
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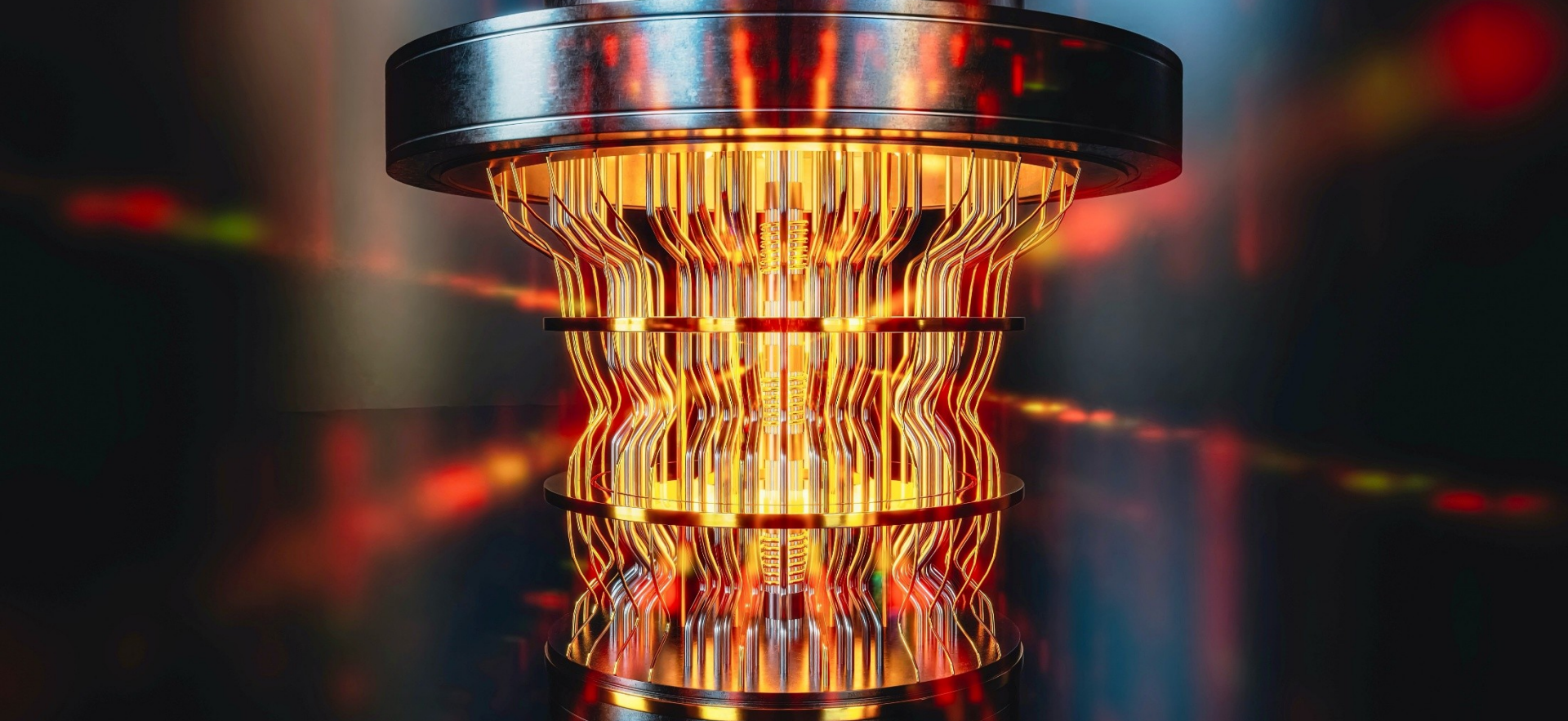
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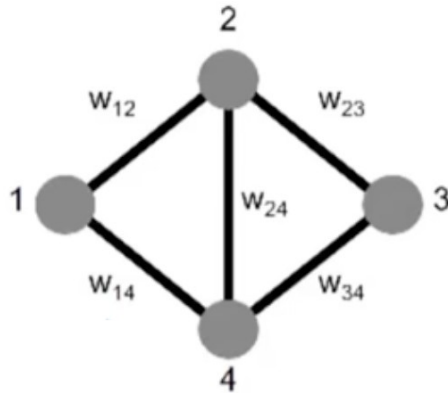
Arbitrary errors (or single qubit transformation) can be decomposed into random applications of Pauli-X and Pauli-Z, and they can be corrected by a clever combination of the two codes, one in the standard basis and one in the e basis, with 8 auxiliary qubits.



Analog Programming with Hamiltonians

::... Introduction

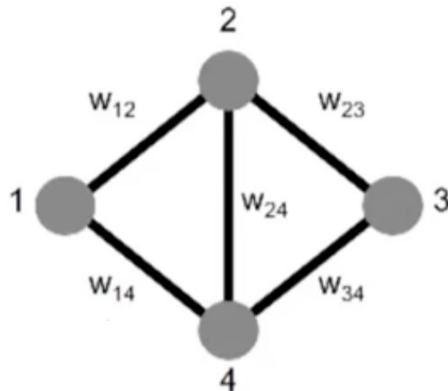
Let us consider a class of minimization problems in which the objective function is to be minimized. An example is the maxcut problem:



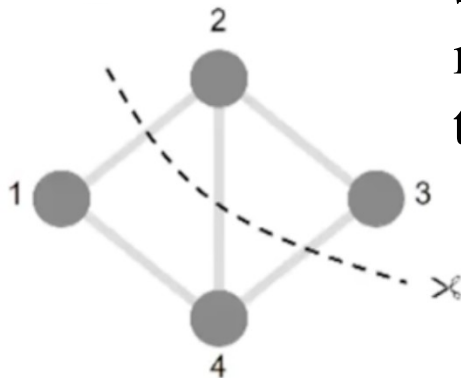
There is a graph G with nodes and edges, where some nodes are connected by the given edges. Each edge carries a weight w_{ij} where i is the starting node and j the arriving one. All weights are positive and .

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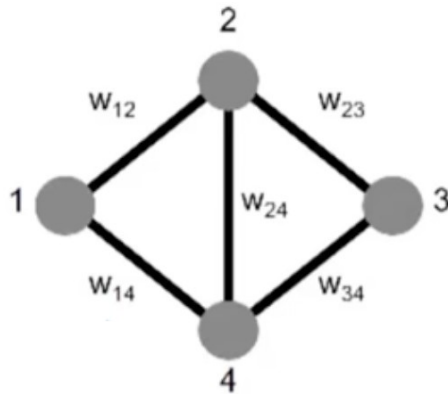
There is a graph G with nodes and edges, where some nodes are connected by the given edges. Each edge carries a weight w_{ij} where i is the starting node and j the arriving one. All weights are positive and $w_{ij} = w_{ji}$.



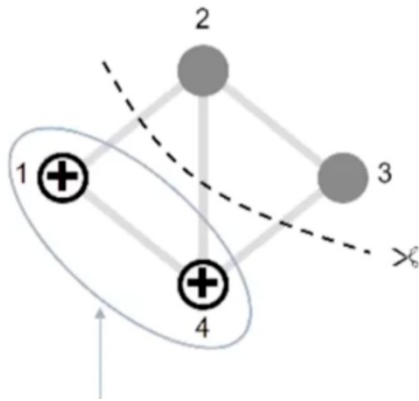
Such a graph can be cut in two parts, and there are multiple possible cut, and each cut separate the nodes in two groups.

::: Introduction

Let us consider a class of minimization problems in which the objective function is to be minimized. An example is the maxcut problem:



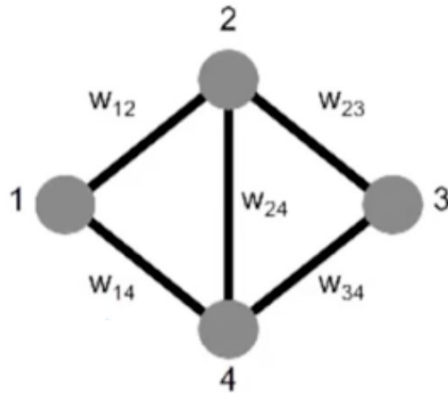
There is a graph G with nodes and edges, where some nodes are connected by the given edges. Each edge carries a weight w_{ij} where i is the starting node and j the arriving one. All weights are positive and $w_{ij} = w_{ji}$.



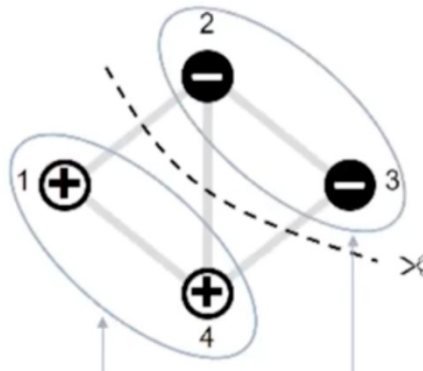
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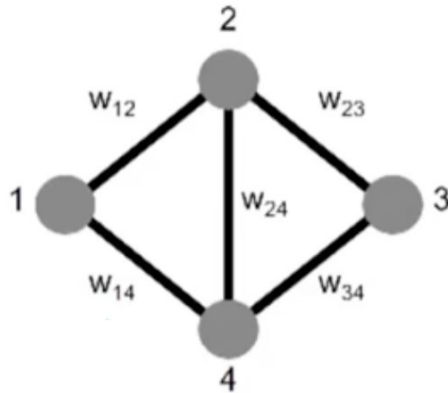
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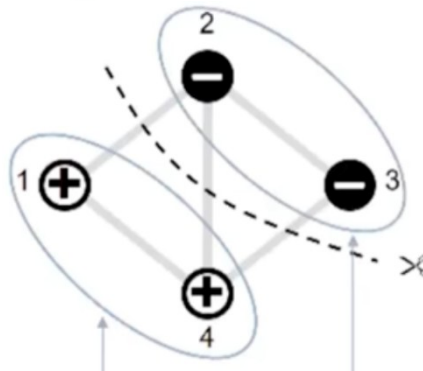
Such a graph can be cut into two parts, and there are multiple possible cuts, and each cut separates the nodes in two groups. We can label each i -th node with a binary variable assigning x_i if the node belongs to the first subset, and $1 - x_i$ if the node belongs to the second subset.

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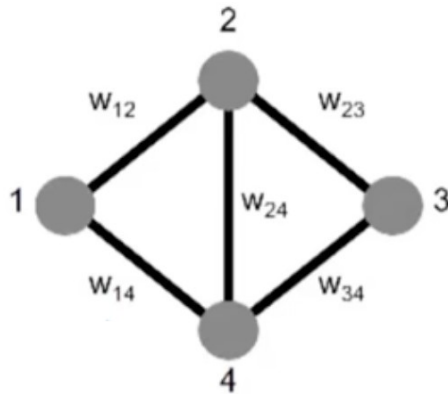
Such a graph can be cut into two parts, and there are multiple possible cuts, and each cut separates the nodes in two groups. We can label each i -th node with a binary variable z_i assigning $z_i = 1$ if the node belongs to the first subset, and $z_i = -1$ if the node belongs to the second subset.

$$\frac{1 - z_i z_j}{2} = \begin{cases} 0 & i, j \text{ in the same subset} \\ 1 & i, j \text{ in different subset} \end{cases}$$

We can define an indicator to help determine if two distinct nodes are in the same subset or not, and which connections were cut or not.

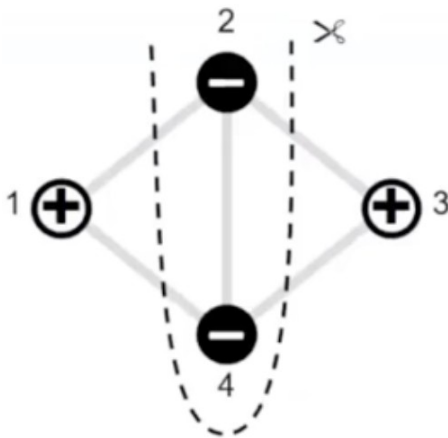
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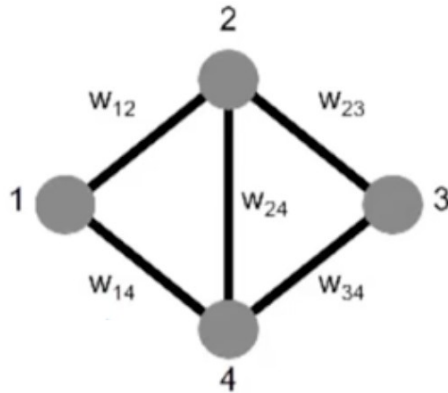
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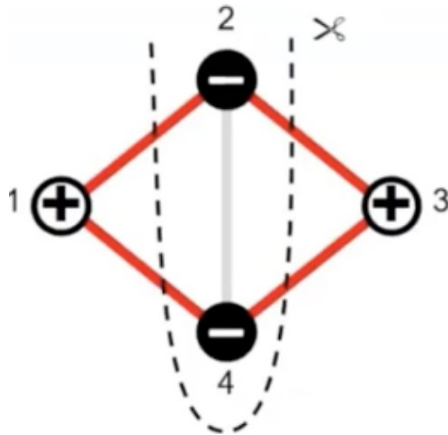


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Here's another possible cut and we assign a value to it: the sum of the edges broken by the cut. The goal is to maximize this value. Classically, this is a hard problem. There is no resolution algorithm to the maxcut problem requiring resources that does not scale exponentially with the number of nodes.

The weighted max-cut problem can be trivially transformed into a weighted minimum cut problem by flipping the sign in all weights.

::... Quantum Analog Computation

We can frame such a problem as one of finding the ground states of an assembly of qubits. Let us formulate the objective of the maxcut problem:

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Maximizing the value of the cut, is equivalent to minimizing the second contribution to the objective function. Many optimization problems can be framed in a cost function to be

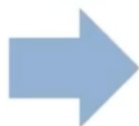
minimised. Now we can pass to the qubit-based formulation: each node is mapped to a qubit, whose state indicates to which subset belongs.

node in subgroup

$$\begin{cases} z = +1 \\ z = -1 \end{cases}$$

cost function

$$C(\vec{z}) := \sum_{i < k} w_{ik} z_i z_k$$



qubit in basis state

$$\begin{cases} |0\rangle \\ |1\rangle \end{cases}$$

Hamiltonian

$$\hat{H}_{MC} := \sum_{i < k} w_{ik} \hat{Z}_i \hat{Z}_k$$

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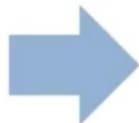
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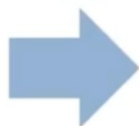
The decision variable has become the Pauli-Z of the i-th qubit.

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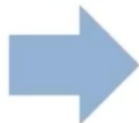
What is an Hamiltonian?

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::... Quantum Analog Computation

The Hamiltonian is the mathematical object that describes the total energy of the quantum system and determines how the system evolves over time. Once the Hamiltonian is known, the change of the quantum state is governed by the Schrödinger equation,

In the Hamiltonian formulation of classical mechanics, a particle is described by a definite position and momentum, which evolve in time according to Hamilton's equations:

The Hamiltonian is the generator of time translations for the position and momentum of a particle.

::... Quantum Analog Computation

Schrödinger's equation has only a first-order time derivative. This means that if we know the initial wave function $\psi(x, 0)$, Schrödinger's equation uniquely determines the wave function for $\psi(x, t)$.

Given two solutions ψ_1 and ψ_2 , any linear combination $c_1\psi_1 + c_2\psi_2$ with constants c_1, c_2 is another solution. This is known as the “principle of superposition”.

The average total energy of a state ψ is

This equation says that the Hamiltonian H tells the quantum state ψ how to change at every moment in time.

If the Hamiltonian is constant in time, the state evolves as

So the Hamiltonian generates the operator $U(t)$, which moves the system from its initial state to its later state.

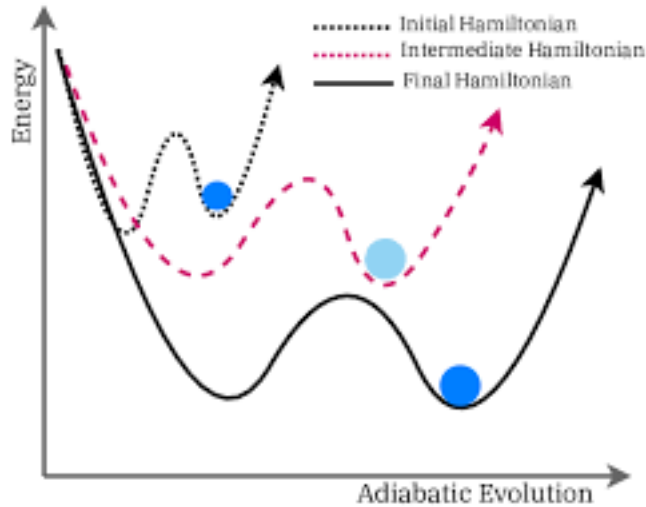
::... Quantum Analog Computation

When we evaluate the expectation value of the Hamiltonian, and every qubit in the basis states will give eigenvalue ϵ_i , while every qubit in the basis states will give eigenvalue ϵ_j . The composite state of all the qubits is an eigenstate of this Hamiltonian, and the eigenvalue is exactly equal to the cost function.

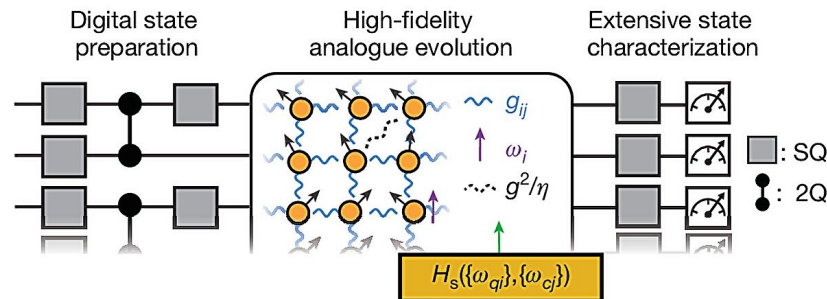
The optimization problem consists of finding the value of the decision variables such that the cost function assumes the lowest possible value, which, in the quantum counterpart, corresponds to finding the basis states in this multi-qubit system that minimizes the energy, corresponding to the eigenvalue of the Hamiltonian.

This is the concept of quantum analog computation, where a physical quantum system solves a problem through its natural continuous-time evolution, rather than through a sequence of discrete logic gates. The experimenter designs and controls the system's Hamiltonian so that the quantum state evolves in a way that encodes the desired computation.

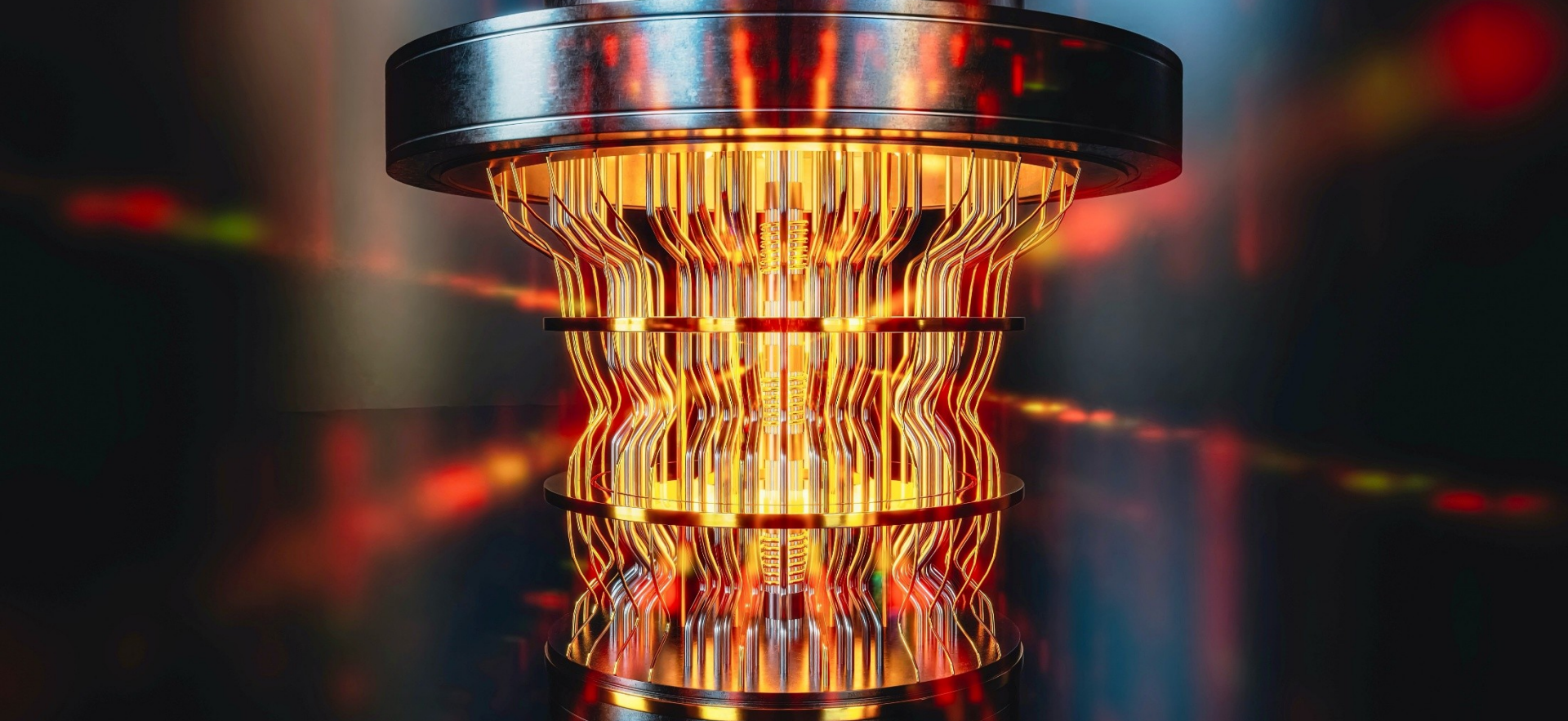
::... Quantum Analog Computation



In quantum annealing, the system is prepared in the ground state of a simple initial Hamiltonian whose ground state is known, and then slowly transformed into a final Hamiltonian whose ground state encodes the solution to the problem. If the evolution is slow enough, the system ideally remains close to the ground state and ends in the answer.

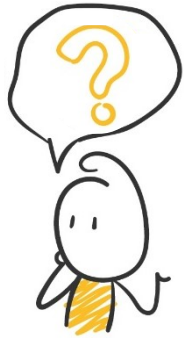


Quantum analog computation is attractive because it can be more natural for simulating physical systems and optimization problems. However, it is also sensitive to noise, control errors, and decoherence, so error correction and precise Hamiltonian engineering are very important.



Quantum Optimization

::... Variational Quantum Eigensolver



What is the most generic form of the n -qubit Hamiltonian?

::... Variational Quantum Eigensolver

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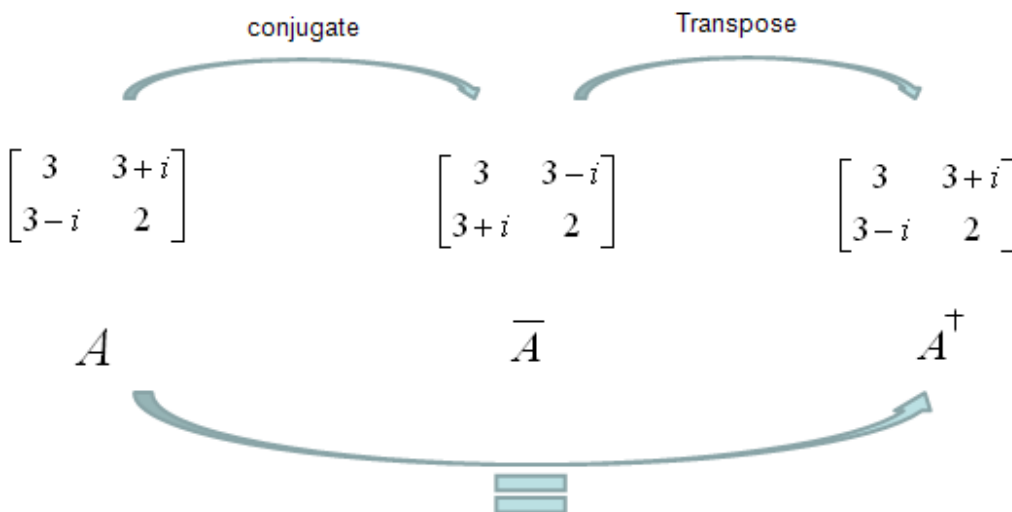
For a single qubit, consider an arbitrary observable in the standard basis, it can be represented by a Hermitian two-by-two matrix, specified by four real numbers.



::... Variational Quantum Eigensolver

What is the most generic form of the n-qubit Hamiltonian?

For a single qubit, consider an arbitrary observable in the standard basis; it can be represented by a **Hermitian two-by-two matrix**, specified by four real numbers.



A 2x2 Hermitian matrix is a square complex matrix equal to its own conjugate transpose ($A^\dagger$), where diagonal elements are real and off-diagonal elements are complex conjugates of each other. The general form is with a, b, c being real numbers.

::... Variational Quantum Eigensolver

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A Hermitian operator is a linear operator characterized by a Hermitian matrix, guaranteeing real eigenvalues and orthogonal eigenvectors.

An operator is Hermitian if it satisfies the condition $\langle \psi | A | \phi \rangle = \langle \phi | A | \psi \rangle^*$ for every state $|\psi\rangle$ and $|\phi\rangle$.

The eigenvalues λ and eigenvectors $|\psi\rangle$ of a linear operator A are nonzero scalars and vectors that satisfy the equation $A|\psi\rangle = \lambda|\psi\rangle$. The operator acts on the eigenvector by changing only its scale (lengthening/shortening) or direction, without rotating it. They are found by calculating the characteristic polynomial $P(\lambda) = \det(A - \lambda I)$.

::... Variational Quantum Eigensolver

What is the most generic form of the n -qubit Hamiltonian?

For a single qubit, consider an arbitrary observable in the standard basis; it can be represented by a Hermitian two-by-two matrix, specified by four real numbers.

Hermitian (or self-adjoint) operators always have real eigenvalues and mutually orthogonal eigenvectors, associated with distinct eigenvalues.

- **Real Eigenvalues:** If $\hat{H}$ is a Hermitian operator, then $\hat{H} = \hat{H}^\dagger$ implies that E is real. This is shown by noting that the expectation value is equal to its complex conjugate.
- **Orthogonal Eigenvectors:** If a Hermitian operator has two distinct eigenvalues with their respective eigenvectors $|\psi_1\rangle$ and $|\psi_2\rangle$, then these eigenvectors are orthogonal, that is $\langle \psi_1 | \psi_2 \rangle = 0$.
- **Complete Basis:** The eigenvectors of a Hermitian operator form a complete orthonormal basis for the Hilbert space.

::... Variational Quantum Eigensolver

What is the most generic form of the n -qubit Hamiltonian?

For a single qubit, consider an arbitrary observable in the standard basis; it can be represented by a Hermitian two-by-two matrix, specified by four real numbers.

Due to two crucial properties, a hermitian operator represents an observable quantity in quantum mechanics, such as energy, spin, or polarization. Specifically, it is the mathematical form of a physically measurable transformation or quantity for a two-state system, such as a qubit, as it guarantees that measurement results are real and that the corresponding states are well-behaved.

Any hermitian matrix can be written as

::... Variational Quantum Eigensolver

What is the most generic form of the n -qubit Hamiltonian?

For a single qubit, consider an arbitrary observable in the standard basis; it can be represented by a Hermitian two-by-two matrix, specified by four real numbers.

The hermitian operators for a single qubit form a four-dimensional linear space, with a basis made of four hermitian operators that are linearly independent. One particular basis is comprised of the three Pauli operators plus the unit matrix. So that, any observable of a single qubit can always be written as a real linear combination of the three Pauli operators and the unit operator, and this can be generalised to the case of many qubits.

::... Variational Quantum Eigensolver

What is the most generic form of the n-qubit Hamiltonian?

For a single qubit, consider an arbitrary observable in the standard basis. It is a Hermitian two-by-two matrix,

The three Pauli operators are the Hermitian matrices

- σ_x : flips $|0\rangle$ and $|1\rangle$;
- σ_y : flips them, but introduces a phase;
- σ_z : leaves the basis states in place, but changes the sign of $|1\rangle$.

They satisfy:

where

any four-dimensional linear operator can be expressed as a linear combination of the **three Pauli**

operators. For a single qubit, the three Pauli operators form a basis. This extends to the case of many qubits.

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::... Variational Quantum Eigensolver

What is the most generic form of the n-qubit Hamiltonian?

For a single qubit, consider an arbitrary observable in the standard basis; it can be represented by a Hermitian two-by-two matrix, specified by four real numbers.

The Schrödinger Hamiltonian operator ($\hat{H}$) in quantum mechanics represents the total energy of a quantum system, adding the kinetic and potential energies. Since measurements of energy must be real numbers, the Hamiltonian must be Hermitian, but not vice versa.

$$\hat{H} = \hat{T} + \hat{V}$$

Kinetic Energy: $\hat{T} = -\frac{\hbar^2}{2} \sum_{k=1}^N \frac{1}{m_k} \nabla^2$

Potential Energy: $\hat{V} = V(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t)$

- Hermitian operator = any operator representing a observable representing a measurable physical quantity;
- Hamiltonian = the specific Hermitian operator representing total energy and generating time evolution.

::... Variational Quantum Eigensolver

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The diagram shows the Schrödinger equation $i\hbar \frac{\partial}{\partial t} \Psi = \hat{H} \Psi$ with several annotations:

- i : square root of minus one (in red)
- $\hbar$: Planck's constant (in orange)
- $\frac{\partial}{\partial t}$: rate of change (in purple) and with respect to time (in green)
- Ψ : quantum wavefunction (in red)
- $\hat{H}$: Hamiltonian operator (in blue)

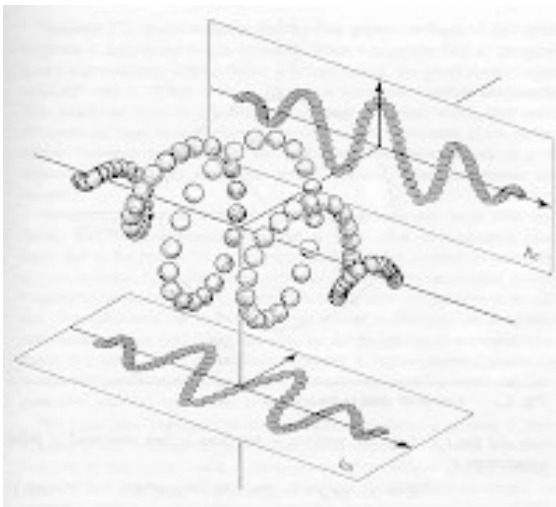
It determines the time evolution of the quantum state, and is central to determining the stationary energy levels and dynamics of quantum systems.

::... Variational Quantum Eigensolver

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The Schrödinger Hamiltonian operator ($\hat{H}$) in quantum mechanics represents the total energy of a quantum system, adding the kinetic and potential energies.



The state of the particle can be prepared so that its state (or wave function) is initially known, and it will then evolve in time according to the Schrodinger equation or a given Hamiltonian operator.

::... Variational Quantum Eigensolver

If we have n qubits described in an n -qubit Hilbert space, then an arbitrary observable in this space can be written again as a real linear combination, and in this case as tensor products, of Pauli operators and unity operators (indicated with σ_i for the Pauli operators and I_i for the unity ones) with real coefficients :

constitute a basis, called the Pauli basis, in the linear space of observables for n qubits.

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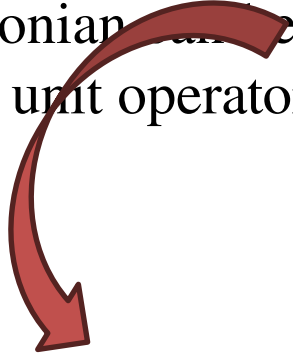
Let's see an example with an Hamiltonian of two qubits with the following matrix representation in the standard basis in the Hilbert space of 2 qubits:

::... Variational Quantum Eigensolver

This Hamiltonian can be written as a linear combination of tensor products of Pauli and unit operators:

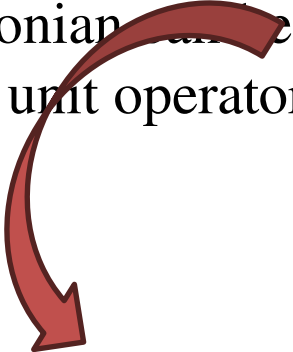
::... Variational Quantum Eigensolver

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::... Variational Quantum Eigensolver

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If we go back to the problem of finding the ground state, we have to try out various states in the Hilbert space and determine the one that minimize the expected value of the energy when the Hamiltonian is used to describe the time evolution of the state.

A procedure to estimate the energy of trial states, makes various measures and then identify the state with the minimum expected energy.

::... Variational Quantum Eigensolver

For any trial two-qubit state $|\psi\rangle$, the energy is $\langle\psi|H|\psi\rangle$, so $\langle\psi|H|\psi\rangle$ have to be measured.

::... Variational Quantum Eigensolver



For any trial two-qubit state $|\psi\rangle$, the energy is $E = \langle \psi | H | \psi \rangle$, so E and $\langle \psi | H | \psi \rangle$ have to be measured.

On a two-qubit state, H does nothing at all:

It does not need a measurement, as it does not distinguish any state from any other. If the observable H is “measured”, the outcome is always E . This is because every state is an eigenstate of H with eigenvalue E . The term E is just a constant energy offset, shifting all energy levels by the same amount, but it does not affect which state is the ground state, making no discrimination between states.

::... Variational Quantum Eigensolver

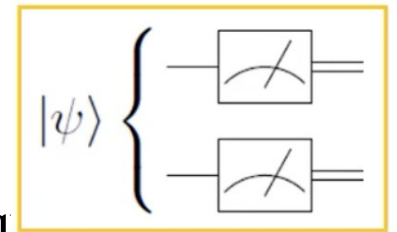
For any trial two-qubit state , the energy is , so

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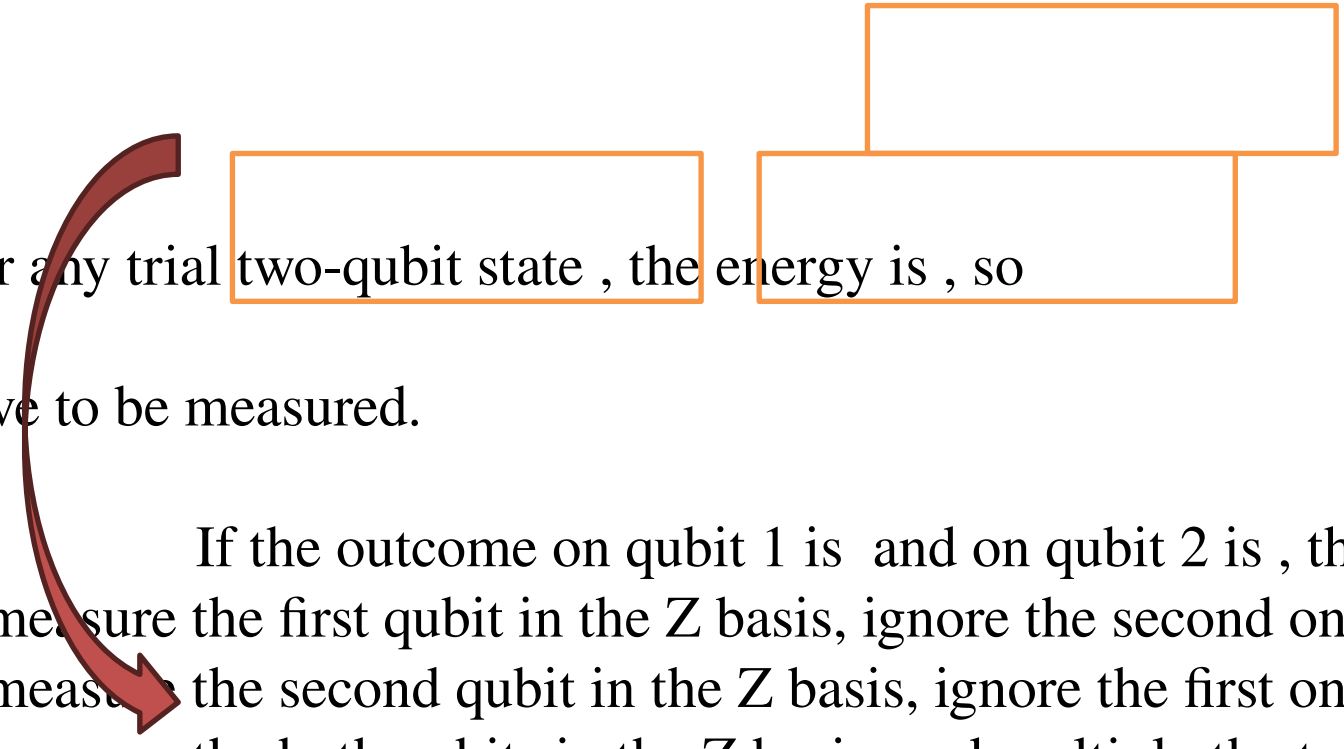
The observables can all be measured in the computational basis. That means:

- prepare the state ,
- measure both qubits in the basis,
- repeat many times,
- estimate the averages from the outcomes.

Since has eigenvalues , the measurement output is if the q if the qubit is found in .



::... Variational Quantum Eigensolver



For any trial two-qubit state , the energy is , so
have to be measured.

If the outcome on qubit 1 is and on qubit 2 is , then:

- measure the first qubit in the Z basis, ignore the second one.
- measure the second qubit in the Z basis, ignore the first one.
- measure the both qubits in the Z basis, and multiply the two outcomes.

::... Variational Quantum Eigensolver

For any trial two-qubit state , the energy is , so

have to be measured.

If the post-rotation measurement outcomes are , then

It measures both qubits in the X basis and multiply the two outcomes. checks whether the two qubits are correlated in the X basis. So unlike , this operator is not diagonal in the computational basis. That is why you cannot read it directly from ordinary Z-basis measurement.



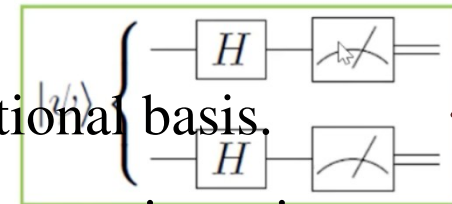
::... Variational Quantum Eigensolver

For any trial two-qubit state , the energy is , so

have to be measured.

The observables can be measured in the Hadamard basis. That means:

- prepare
- apply Hadamard on qubit 1 and 2,
- measure both qubits in the computational basis.



Because Hadamard changes basis as , so measuring is converted into measuring after the Hadamard rotation.

::... Variational Quantum Eigensolver

Take

Then

So the Hamiltonian is

::... Variational Quantum Eigensolver

In the computational basis

this becomes

The diagonal part alone would give energies:

So among basis states, $|0\rangle$ is best. But the term $H_{\text{off-diag}}$ mixes states and may lower the energy further.

::... Variational Quantum Eigensolver

Because $|00\rangle$ and $|11\rangle$, since $\sigma_z \otimes \sigma_z$, applying to both qubits gives
and

The same happens for the other pair:
and

That is why we say it “couples” those states.

For

the expectation values needed for the energy are computed. The Hamiltonian contains $\sigma_x \otimes \sigma_x$, and since the term has coefficient J . So

::... Variational Quantum Eigensolver

We get:

a) Expectation of

b) Expectation of

for every .

c) Expectation of

Thus, the energy is

so

::... Variational Quantum Eigensolver

Case 1:

Case 2:

This is lower than , so the superposition is better than .

Case 3:

Again not as good as .

::... Variational Quantum Eigensolver

For the measurement-based approach to estimate the ground-state energy of

the circuit has two parts:

- a state-preparation circuit for a trial state ;
- a measurement circuit adapted to each Pauli term.

::... Variational Quantum Eigensolver

For the measurement-based approach to estimate the ground-state energy of the circuit has two parts:

- **a state-preparation circuit for a trial state ;**
- a measurement circuit adapted to each Pauli term.

One simple way to prepare an entangled two-qubit trial state is to start from $|00\rangle$, rotate qubit 1, entangle with C-NOT and optionally add single-qubit phases/rotations.

This prepares the trial state .



::... Variational Quantum Eigensolver

For the measurement-based approach to estimate the ground-state energy of the circuit has two parts:

- **a state-preparation circuit for a trial state ;**
- a measurement circuit adapted to each Pauli term.

An **ansatz** is a trial form for the quantum state, written with some adjustable parameters to be varies so as to approximate the true solution of the problem, such as the ground state of a Hamiltonian.

Instead of searching over all possible states in the Hilbert space, which is too hard, it is possible to choose a family of states like

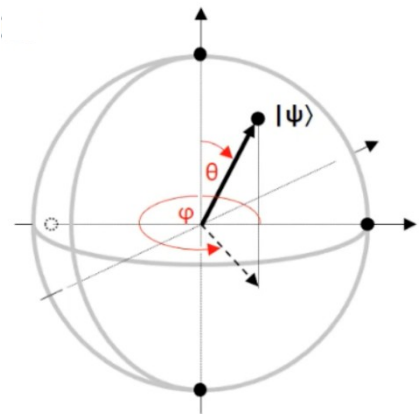
and then optimize the parameters based on a given objective.

An ansatz is often implemented as a parametrized quantum circuit.

::... Variational Quantum Eigensolver

For the measurement-based approach to estimate the ground-state energy of the circuit has two parts:

- **a state-preparation circuit for a trial state ;**
- a measurement circuit adapted to each Pauli term.



The state of a single qubit can be parametrised based of two angles θ and ϕ , and this can be translated in the language of gates and circuits as follows:

es for rot $|0\rangle \xrightarrow{R_y(\theta)} \xrightarrow{R_z(\phi)} |\psi(\theta, \phi)\rangle$ a given

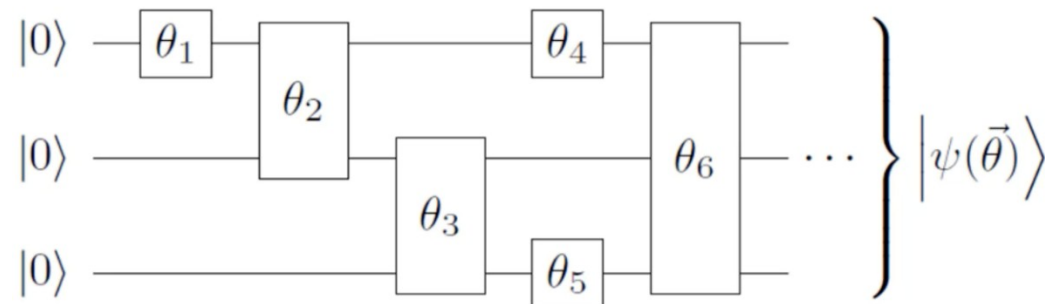
axis for a give angle. This circuit allows to prepare any single qubit state with a fixed set and types of gates, but varying the parameters.

::... Variational Quantum Eigensolver

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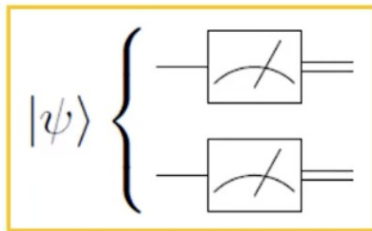
From single qubit state to multi-qubit states, we can generalise this idea by parametrizing many-qubit state by . We can think of a certain fixed architecture for any state to be prepared, but the specific parameters varies.



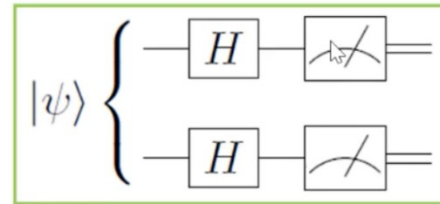
::... Variational Quantum Eigensolver

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and



::... Variational Quantum Eigensolver

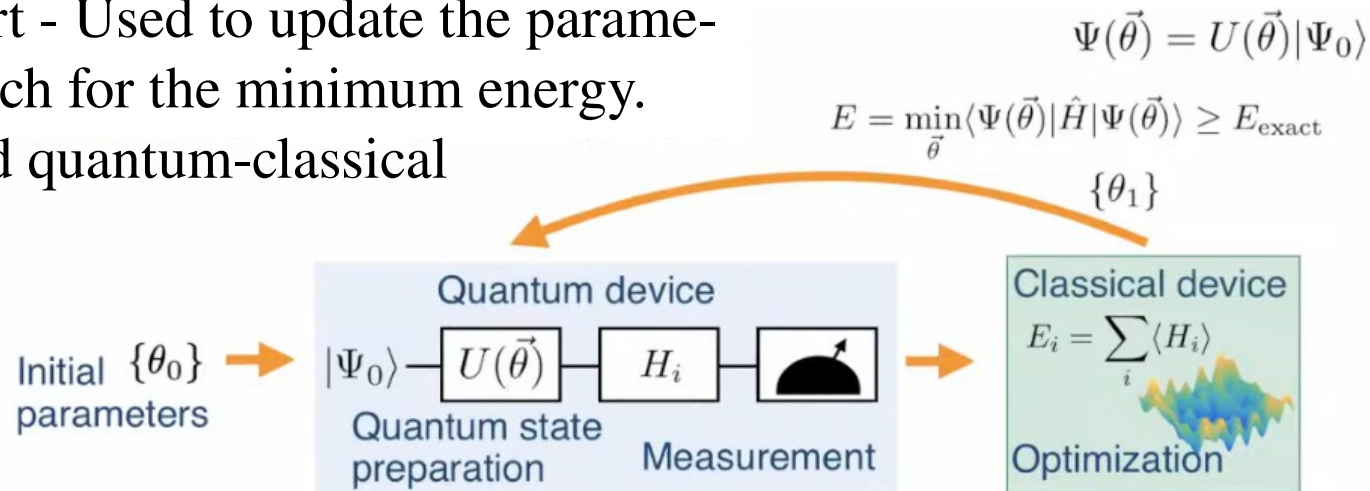
The **Variational Quantum Eigensolver** (VQE) is exactly the method that realizes this idea. and is based on the variational principle:

for every normalized state $|\Psi(\vec{\theta})\rangle$, where E_{exact} is the true ground-state energy. So by searching over many trial states, the one with the lowest measured energy is your best approximation to the ground state.

A VQE has two parts:

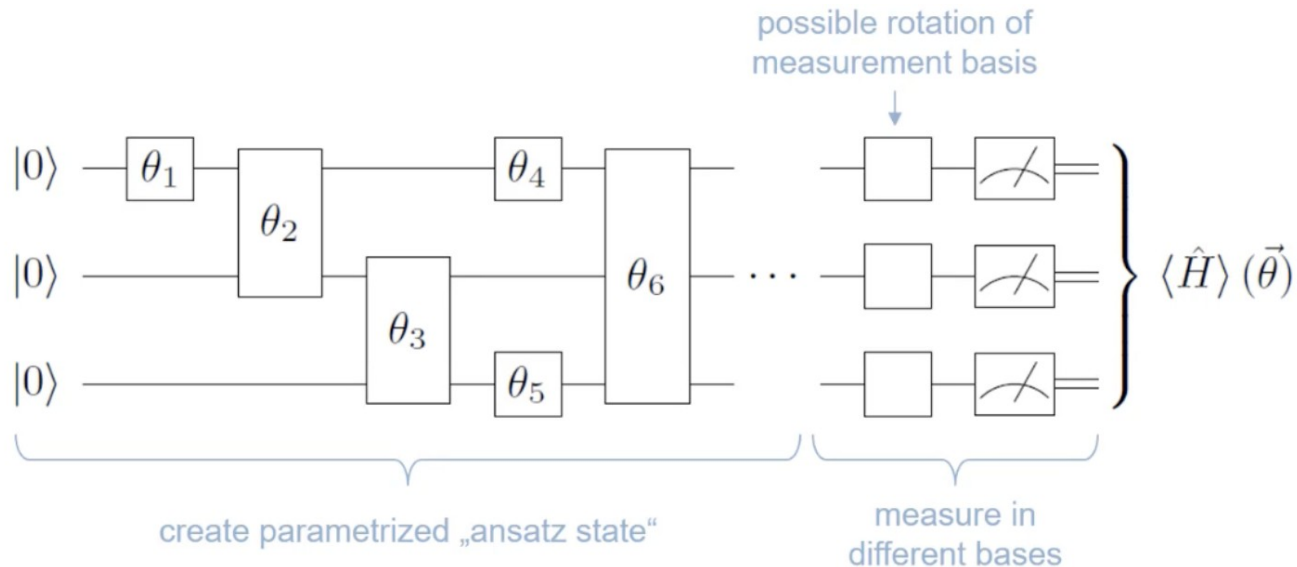
1. Quantum part - Used to prepare the ansatz and measure the Hamiltonian terms.
2. Classical part - Used to update the parameters and search for the minimum energy.

So VQE is a hybrid quantum-classical algorithm.



::... Variational Quantum Eigensolver

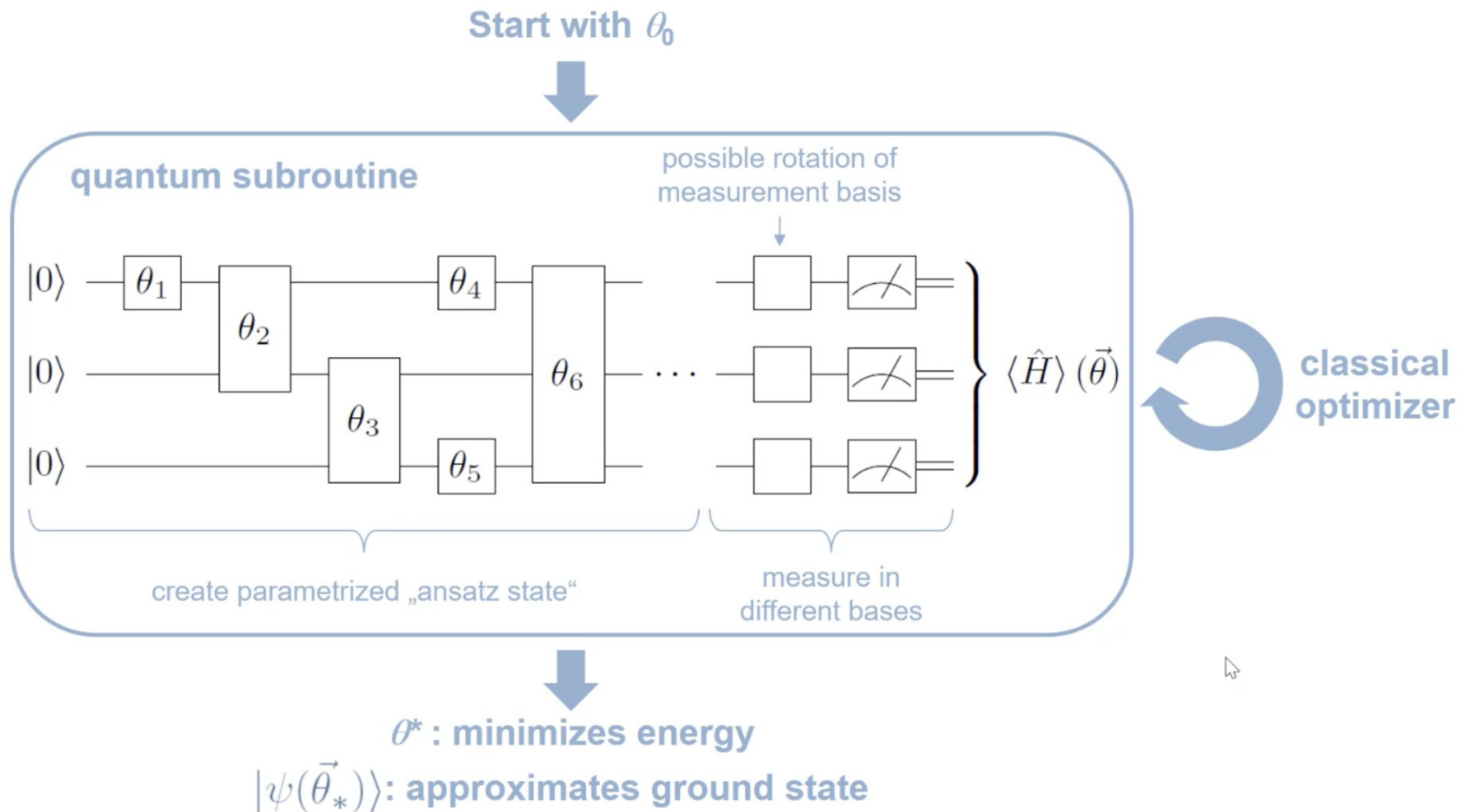
1. Quantum part - Used to prepare the ansatz and measure the Hamiltonian terms.



The expectation value of the Hamiltonian (aka the energy of the system) is function of the set of parameters used in the ansatz to prepare the state .

::... Variational Quantum Eigensolver

2. Classical part - Used to update the parameters and search for the minimum energy.



::... Variational Quantum Eigensolver

2. Classical part - Used to update the parameters and search for the minimum energy

The optimizer does not solve the physics directly. Its role is to solve the classical numerical minimization problem, where the function is evaluated by the quantum processor through repeated measurements.

- Derivative-free optimizers are very common, especially on noisy hardware. A standard example is COBYLA, a constrained optimization method that does not require derivatives.
- Gradient-based optimizers can also be used when gradients are available or can be estimated.
- Noise-tolerant stochastic optimizers are often preferred on hardware. A common example is SPSA, whose key feature is a stochastic gradient approximation requiring only two objective-function evaluations regardless of the number of parameters.

In code, it is realized as a standard classical software component, usually running on a CPU.

$\hat{H}(\vec{\theta})$



::... Variational Quantum Eigensolver

In principle, to find the ground-state energy, all the possible states in the n -qubit Hilbert space has to be tried out. As the complexity grows exponentially with the number of qubits, this approach is not efficient.

The secret to have a more efficient solution is to include in the ansatz two-qubit gates that create entanglement between the qubits. The circuit for the state preparation create highly entangled ansatz states, which are superpositions of many different basis states of the n qubit and used to measure expectation energy values.

The algorithm is effectively exploring many basis states simultaneously, and its number of parameters does not scale exponentially with the number of qubits, but it scales linearly.

::... Variational Quantum Eigensolver

VQE is one of the main algorithms designed to be run on current NISQ quantum devices, meaning quantum computers with limited qubit counts and imperfect gates, by splitting the work between a quantum processor and a classical optimizer. This is in contrast to Quantum Phase Estimation, which generally requires fault-tolerant machines and deeper circuits.

VQE uses shallower parametrized circuits and repeated expectation-value measurements, rather than long coherent computations. VQE relies on repeated measurement estimation, which is exactly the kind of workload current hardware can perform.

That said, “realizable on current devices” does not mean “already broadly superior in practice.” VQE is implementable today for small problems and proof-of-concept experiments, but it remains sensitive to noise, shot cost, ansatz choice, and optimization difficulties.

::... VQE applied to QUBO

A QUBO optimization can be solved with VQE by turning the QUBO cost function into a qubit Hamiltonian (represented as Pauli operators) and then using VQE to approximate that Hamiltonian's ground state.

1) Start from the QUBO

A QUBO problem is usually written as

or compactly as $\sum_i c_i x_i$, with binary variables $x_i \in \{0, 1\}$.

2) Map binary variables to qubit operators

To solve the QUBO with VQE, you convert each binary variable x_i into a qubit operator. The standard mapping is

where σ_i^x is the Pauli- x operator acting on qubit i .

::... VQE applied to QUBO

After substitution, the QUBO becomes a Hamiltonian of the form

This is an Ising Hamiltonian. The important fact is that the bitstring that minimizes the original QUBO corresponds to the computational-basis state whose energy under is minimal. So the QUBO optimum becomes the ground-state problem of a Hermitian operator.

3) Why the ground state solves the optimization problem

Once the QUBO is converted into a Hamiltonian, each computational-basis state represents one candidate binary solution. Because the Hamiltonian is constructed from z -type terms, it is diagonal in the computational basis, and its diagonal entries are exactly the QUBO objective values up to a constant offset. Therefore,

where E is the cost associated with bitstring $\mathbf{z}$.

::... VQE applied to QUBO

The bitstring with lowest cost is the basis state with the lowest eigenvalue, that is, the ground state. So the optimization problem becomes:

That is the conceptual core of solving QUBO with VQE.

4) What VQE does with that Hamiltonian

VQE does not search directly over all bitstrings. Instead, it chooses a parameterized quantum circuit (the ansatz) that prepares a family of states

where θ is a vector of tunable parameters. The quantum computer estimates

A classical optimizer then changes θ to reduce $E(\theta)$. If VQE finds parameters with very low expectation value, the corresponding state is a good approximation to the ground state, and measuring that state gives candidate solutions to the original QUBO.

::... VQE applied to QUBO

5) How the energy is measured term by term

Because the Hamiltonian is written as a sum of Pauli strings,

the energy expectation is

So VQE estimates each Pauli expectation value separately from repeated measurements and then reconstructs the total energy as a weighted sum. For a QUBO-derived Ising Hamiltonian, the terms are usually only σ_z and $\sigma_z \sigma_z$. That is convenient because they are all diagonal in the computational basis and can often be measured in the same basis setting. This makes QUBO-to-VQE conceptually simpler than Hamiltonians with mixed σ_x , σ_y , and σ_z terms.

::... VQE applied to QUBO

6) What the classical optimizer is minimizing

The classical part of VQE solves the numerical problem

COBYLA, SPSA, L-BFGS-B, or another routine decide which parameter vector to try next; the quantum processor supplies the energy estimates. At convergence, the best parameters define the VQE approximation to the ground state.

7) How the binary answer is extracted at the end

After VQE has found a low-energy state, the bitstring solution to the original QUBO is still needed. The most likely bitstring is taken as the candidate optimum. If the ground state is degenerate or the ansatz produces a superposition over several low-cost states, several high-probability bitstrings can be seen.

::... VQE applied to QUBO

Suppose

Using

you get a Hamiltonian of the form

VQE prepares a trial state ψ , measures $\langle H \rangle$, and σ , reconstructs H , and optimizes ψ . When the energy is minimized, measuring the final state gives the bitstring that minimizes $\langle H \rangle$.

QUBO is a very broad modeling language able to describe many problems; once in QUBO, a given problem can be translated to an Ising Hamiltonian and attacked with quantum optimization routines. VQE only gives an approximation unless the ansatz is expressive enough and the classical optimizer finds a good minimum. On real hardware, shot noise and hardware noise affect the measured energy.

::... QAOA

The **Quantum Approximate Optimization Algorithm** (QAOA) is a hybrid quantum algorithm, designed to find approximate solutions to combinatorial optimization problems on NISQ devices. It holds many aspects in common with VQE, but also many differences

- Both are
 - variational methods,
 - used to minimize an expectation value,
 - hybrid quantum-classical,
 - intended for NISQ devices,
 - made of repeated measurements, characterized by performance depending on how expressive the circuit is, whether the optimizer finds a good minimum, how much noise affects the measurements,
 - understood as low-energy search.
- Differences are
 - VQE is a general variational framework, but QAOA is a specific structured variational ansatz,

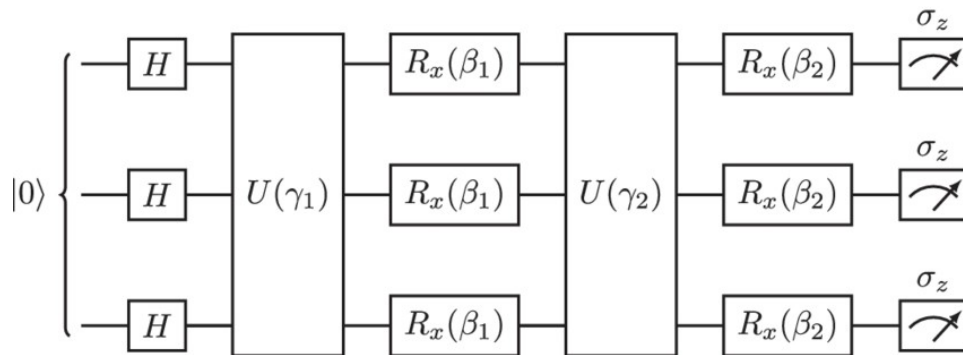
::... QAOA

- 0 In VQE, the ansatz is generally flexible, but in QAOA, the ansatz is much more structured.

QAOA's ansatz alternates between:

- evolution under a cost Hamiltonian ,
- evolution under a mixer Hamiltonian .

At depth , the standard QAOA state is



So, QAOA has main parameters in the standard version and not (the number of variables of the QUBO problem).

Note that the two Hamiltonians must not commute to have not trivial results.

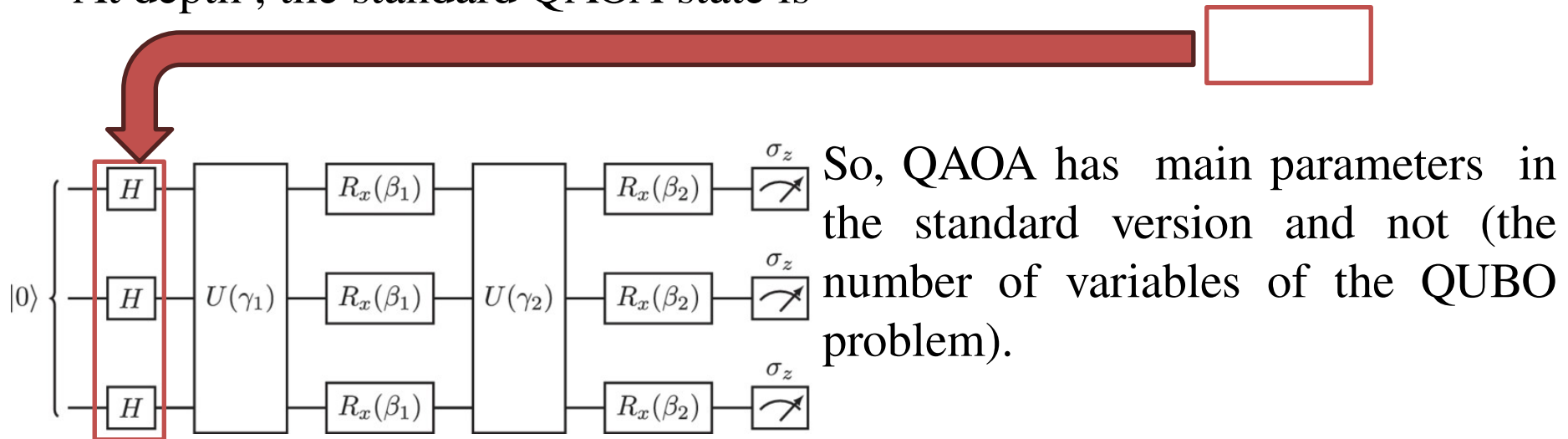
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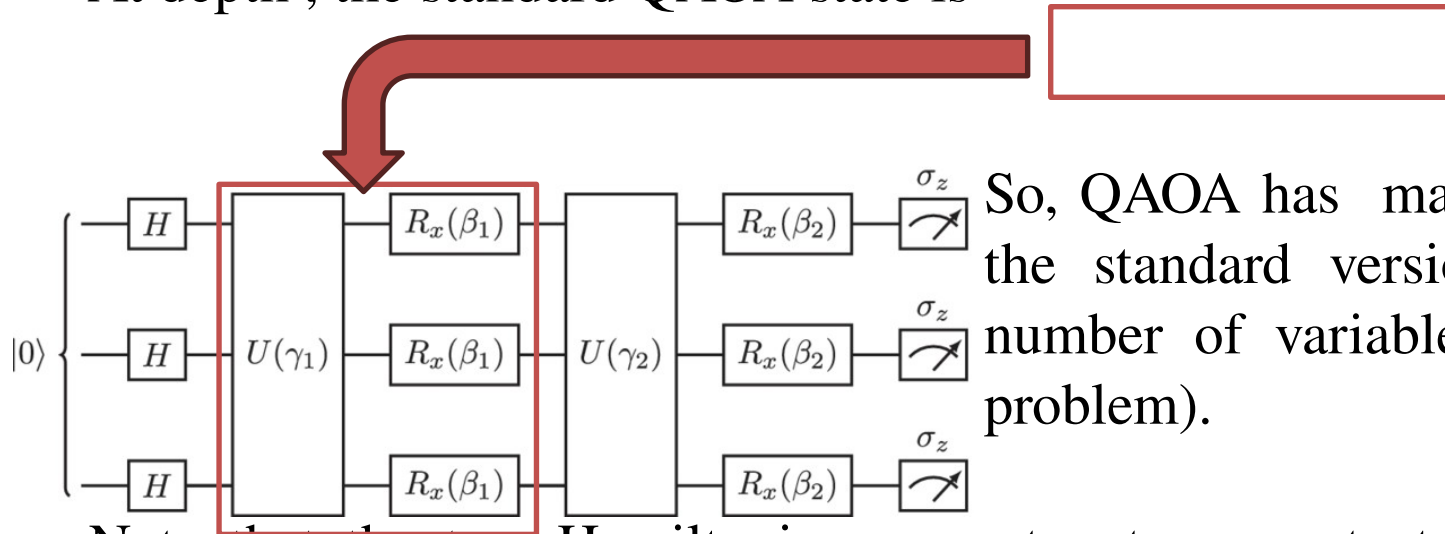
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::... QAOA

- In VQE, the Hamiltonian can be very general, but in QAOA, the cost Hamiltonian is usually derived from a classical objective function, often diagonal in the computational basis.
- In VQE, the main quantity of interest is usually the minimum eigenvalue and sometimes also the approximate ground-state wavefunction. In QAOA, the practical output is usually a high-quality bitstring sampled from the final state, corresponding to a good approximate solution of the combinatorial problem.
- In VQE, the number of parameters depends entirely on the ansatz you choose. In QAOA, the parameterization is much more standardized, and its complexity is controlled very explicitly by p , and the approximation quality can improve as p increases, at the cost of deeper circuits.
- In standard QAOA, one usually starts from $|0\rangle$ and applies alternating cost and mixer unitaries. In VQE, there is no such mandatory starting state or mixer concept.

::... QAOA

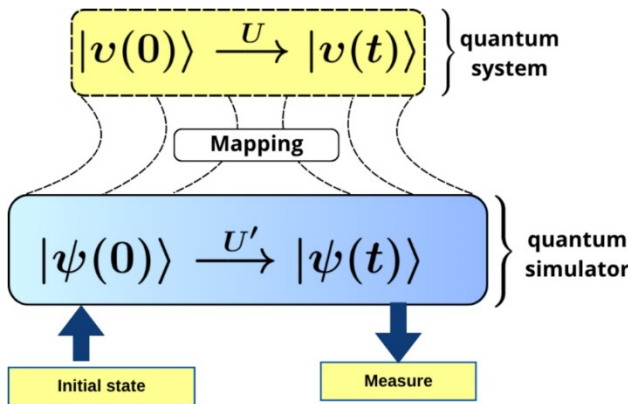
QAOA is often described as a special case or highly structured relative of VQE. Starting from a QUBO problem, you apply QAOA by turning the binary objective into an Ising cost Hamiltonian . The standard QAOA mixer is

where σ_x is the Pauli- on qubit . The mixer's role is to move amplitude among bitstrings so the algorithm can explore the solution space rather than staying in one basis state. At this time, the algorithm is executed, and variational parameters optimised.

After optimization, you run the final QAOA circuit many times and measure in the computational basis. The most likely low-cost bitstrings are your candidate QUBO solutions.

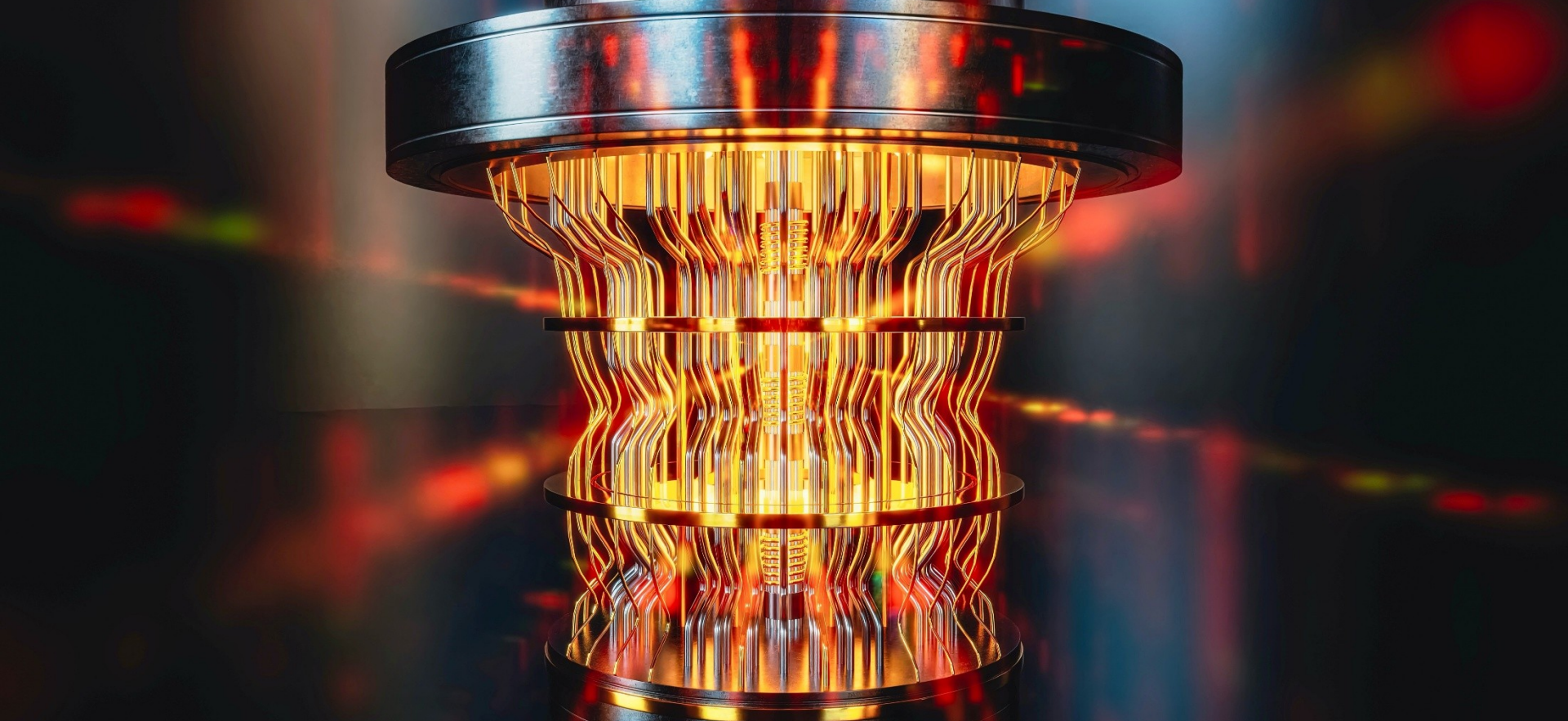
::... Variational Quantum Simulation

The **Variational Quantum Simulation** (VQS) is a family of hybrid quantum-classical algorithms that use a parameterized quantum state or circuit to approximate the time evolution of a quantum system, including real-time and imaginary-time evolution. So, at a high level, VQS is about this idea:



Instead of implementing the exact evolution directly, you choose an ansatz with parameters , and then update those parameters so that the ansatz follows the target dynamics as closely as possible.

VQS usually returns an approximate time-evolved state trajectory or properties derived from it, rather than a static optimization answer.



Quantum Machine Learning